



V J Desai & Co. LLP · GST Keeper

PRODUCT REVIEW · ROADMAP · BLUEPRINT

Notices Dashboard and Litigation Management

A UX and engineering review of the Notices Dashboard, what is broken and missing today, and a phased plan to rebuild it as a litigation ERP that emails the GST team about every matter that needs attention.

PREPARED FOR

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DATE

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SCOPE

gst.vjdesai.com › Notices Dashboard family, portal sync, email pipeline

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STATUS

Draft v1.0 for partner discussion

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Figures: 1 current dashboard (annotated recreation) · 2 redesigned dashboard · 3 matter drawer · 4 daily digest email. All mock-ups are rendered with the app's own design tokens (navy primary, coral accent, Inter/Poppins) and realistic sample data; client names in the mock-ups are illustrative.

1 Executive summary

The Notices Dashboard looks the part but cannot yet be trusted or acted on. It was built as a faithful copy of a competitor's layout; the numbers on it are undermined by one defect in the portal sync, and nothing in the system tells the GST team about a notice unless someone opens the page. This document sets out what is broken, what a litigation ERP for the firm needs, a redesigned dashboard built around a daily work queue, an email alerting layer on the pipeline the firm already runs, and an 18-week phased roadmap.

32

VERIFIED DEFECTS

3 critical · 9 high; every claim survived two adversarial verifiers

31

FURTHER DEFECTS

From the follow-up reviews; the severe ones hand-checked (\$4.2)

123

GAPS AND IMPROVEMENTS

UX, data model, alerting, litigation, sync, security, retention

~130

PERSON-DAYS

6 phases · 20 weeks with one developer, ~12 with two

The seven findings that matter most

- 1. Every Sync All erases the team's work.** The extension deletes all of a client's notice rows and re-inserts what the portal returns (`extension/background.js:102–105`). Status, priority, owner, reply and order details entered by staff are wiped each time. This is why the live page shows Open = Total = 1,060 and a Replied column of dashes. A failed pull is worse: it replaces the client's entire history with one "PULL FAILED" row, which is then counted as a notice.
- 2. The KPI tiles do not mean what they say.** "Last 24 Hours" counts rows re-written by the last sync, not new notices; "Over Due" counts closed and replied items, ignores extended due dates and turns a notice overdue from 05:30 IST on its own due date; DRC-01, ASMT-10 and other case rows are written without any due date, so the items with the most at stake can never appear as due or overdue; three reports use three different definitions of "a notice" and cannot reconcile.
- 3. Nothing emails anyone about a notice.** The firm has a working email pipeline (Graph sender, outbox, editable templates, cron) used only for return-data reminders. There is no event for a new notice, no due-date ladder, no digest, no hearing or appeal-limitation reminder, no failed-login alert, and the outbox schema physically blocks team-level alerts (single recipient, client required, fixed kinds).
- 4. There is no litigation model.** A dispute that runs from DRC-01A to DRC-01 to reply to hearing to DRC-07 to APL-01 is stored as unrelated flat rows with free-text status and free-text assignee. No stages, statutory deadlines, hearings, demand breakdown, pre-deposit tracking, documents repository, activity log or client view exist.
- 5. The page is a set of counts, not a place to work.** Eight tiles, a client-onboarding card and a month calendar sit above a right-hand table of dashes, all in 10–11 px text, with no list of notices, no owner, ageing, amount at stake or "what needs attention today". Below the tiles the screen is empty at 1920 × 1080.
- 6. Sync is manual, browser-bound, blind, and not wired to the production domain.** Fresh data depends on a person in Chrome clicking Sync All and typing a CAPTCHA per client; the extension bridge is injected only on `*.vercel.app`, so on `gst.vjdesai.com` the button reports "Extension not

detected" (as in the screenshot); nothing records what changed; the Windows agent runner exists but has no notices handler and no schedule.

7. **Litigation documents and credentials are not protected.** Every scanned notice and order PDF sits in a public storage bucket at a guessable path and is fetchable without a login; storage policies let anyone with the public key delete or overwrite them; a client-role login can open the Clients page and read every portal password unmasked; the session role is restored from the browser's localStorage copy rather than the database, so a client user can promote itself; and the extension bridge accepts commands from any web page. These were found by the follow-up review and hand-checked by the author. They are not notice-dashboard bugs in the narrow sense, but no alerting layer should email links to those PDFs until they are closed, so they sit in Phase 0.

What we propose

Phase 0 (3 weeks) makes the numbers true and closes the holes. Replace delete-then-insert with an upsert that preserves staff columns, stop failures from writing notice rows, define every tile once, auto-close the acknowledgement rows that inflate "open" today, fix the production-domain and security defects, and write the module's positions down.

Phase 1 (3 weeks) emails the GST team on thirteen events (new notice, due ladder, hearing, appeal limitation, critical recovery / detention notices, assignment, closure, sync failure, daily digest, weekly MIS, unassigned items, and two client-facing messages), all through the existing outbox and sender (figure 4), with a named owner and early statutory clocks so the ladder reaches a person.

Phase 2 (4 weeks) rebuilds the page around a work queue (figure 2): a needs-attention strip, a sortable queue with bulk actions, ageing and exposure charts, a 14-day deadline strip, a sync-and-alerts card, refunds and DRC-03 with real rupee figures, and a matter drawer behind every row (figure 3).

Phase 3 (5 weeks) adds the litigation layer: matters with stages, statutory deadline and pre-deposit engine, hearings, demand and payments linked to DRC-03, documents and a reply workspace, partner review, outcomes and a read-only client view.

Phase 4 (3 weeks) keeps it running: unattended sync through the existing agent runner with session reuse (no CAPTCHA auto-solving), a 06:00 IST schedule, run summaries, a CAPTCHA queue, and retention and recovery so a click or a cascade cannot lose litigation records.

Phase 5 (2 weeks) delivers MIS and client reporting: exposure, ageing, outcomes and per-staff load, a client litigation status PDF on the letterhead, and an in-app notification bell.

Decisions the partners need to make first (detail in §11.4)

(1) Confirm the legal posture for unattended portal access stays "human-typed CAPTCHA, session reuse, written client authorisation". (2) Name the team distribution list and who counts as "partner" for escalations. (3) Confirm the auto-close rules for LUT and DRC-03 acknowledgements. (4) Confirm the statutory defaults in §7.1 as the firm's positions. (5) Approve the Phase 0 security stop-gaps and a separate project-wide security track (credential encryption, Supabase Auth for staff) so RLS can become real.

2 Scope and how this review was done**2.1 What was reviewed**

- The live page as screenshotted on 13 Sep 2026 (1,060 notices, 92 GSTINs) and its source: `src/pages/NoticesDashboardPage.tsx` and the seven sibling pages of the Notices family (all-clients list, company list and profile, case folder, summary and GSTIN-wise reports, refunds and DRC-03 lists).
- Shared helpers: `noticeSummaryReport.ts`, `noticeCategoryClassifier.ts`, `NoticeWorkflowListView.tsx`, `AddNoticeDialog.tsx`, `fetchAllRows.ts`.
- Schema: every migration touching `gst_notices`, refunds, DRC-03, case folders, sync log, portal agent, reminders and email.
- Portal sync: the Chrome extension (`background.js`, `content.js`) and the Windows agent runner (`agent/`).
- Email pipeline: `send-gst-email`, `queue-gst-reminders`, the corporate email shell, cron jobs and the Reminders page.
- Design system rules in `docs/UI-UX-REDESIGN.md`, tokens in `src/index.css` and `tailwind.config.ts`.

Not done, and why

- No query was run against the live Supabase project; counts quoted are from the screenshot and from comments in the code that record live checks. Phase 0 starts with the live measurements listed in §11.1.
- No change was made to the application. Mock-ups are static renders, not the app.
- Competitor features (Appendix B) come from public web sources and are labelled as such.

2.2 Method

The review ran as a structured, multi-reviewer process. 8 independent reviewers each read the code through one lens; every claim that something is *broken* was then handed to two adversarial verifiers whose brief was to refute it from the code (one tracing the actual data flow, one testing realistic data shapes and the soundness of the proposed fix). A claim survived only if it was not refuted by both. A completeness critic then named what the first pass had not covered, and 6 targeted follow-up reviews closed those gaps. Three competing roadmap drafts (fix-first, user-first, platform-first) were scored by two judges, a partner persona and a lead-engineer persona, and the roadmap in §10 is the merge

they recommended. The author separately re-read the decisive files (`background.js` , `content.js` , `fetchAllRows.ts` , the GSTIN-wise report) to confirm the headline defects by hand.

Review lens	What it examined
Look and feel	Hierarchy, density, tokens and type sizes, width use at 1920/1366/phone, tile semantics, affordances, empty/error states, accessibility.
Correctness	Every computation and drill-down link on the page, date handling, dedupe, paging, error handling, sibling-page consistency.
Data model	Tables, keys, constraints, RLS, what a litigation lifecycle needs and has no home.
Email and alerting	Existing outbox, templates, cron, sender; every missing notice event and the schema that blocks it.
Litigation ERP	The CGST lifecycles (scrutiny, audit, demand, appeal, tribunal, recovery, refund, registration, e-way bill, summons, amnesty) against what the app captures.
Portal sync	Extension and agent write paths, failure handling, idempotency, scheduling, observability.
Benchmark	What Indian notice- and litigation-management products offer that this app does not.
Staff workflow	The click path from "a notice landed" to "partner reviewed the reply", across all pages.

3 The dashboard today, annotated

Figure 1 recreates the live page from its source at 1600 px and marks the twelve observations that drive the redesign. The layout is a close copy of the Notice Alert product the firm pointed to when the page was built; that parity was a reasonable first target, but it imported the competitor's weaknesses (count tiles, a permanent table of placeholder rows, a month calendar with no counts) and added one of its own: onboarding buttons on a dashboard that should be about notices.

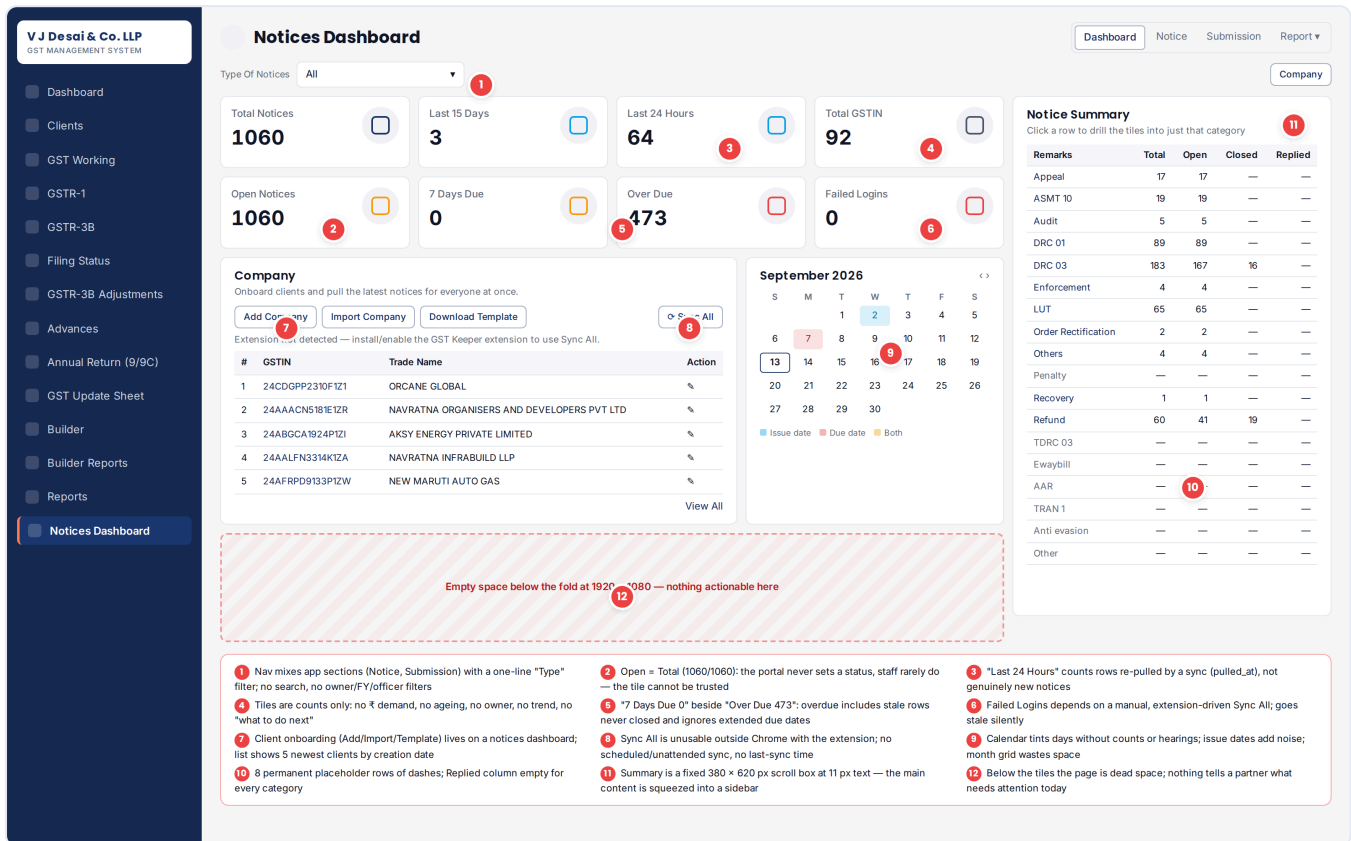


Figure 1. Current Notices Dashboard, recreated from source and annotated. Numbers on the tiles are those in the 13 Sep 2026 screenshot.

What a partner sees. Total 1,060 and Open 1,060 say the firm has never closed a notice. Over Due 473 next to "7 Days Due 0" says nearly half the portfolio is late and nothing is coming up, which no litigation practice would believe. Last 24 Hours 64 against Last 15 Days 3 says more arrived today than in the last fortnight. Each of these is an artefact of how the data is written, not a fact about the practice (§4).

What a staff member cannot do here. See a single notice; see who owns anything; see money at stake; sort by deadline; assign, prioritise or close in bulk; search; or learn when the data was last refreshed.

What the visual system does. Twenty-three uses of 10–11 px text, muted-on-muted tints, icon-only buttons without labels, a fixed 380 × 620 px scroll box holding the main content, and dead space under the second row. The app's own design rules (semantic tokens, 8-point rhythm, PageHeader, TableEmptyState, aria-labels on icon buttons) are documented in docs/UI-UX-REDESIGN.md and are not followed on this page.

What is good and stays. The category taxonomy (Appeal, ASMT 10, DRC 01, DRC 03, LUT, Refund...), the drill-down idea behind every number, the per-client profile and case-folder pages, and the portal PDF capture are sound and are reused throughout the redesign.

Live-data check (read-only, public key, 13 Sep 2026)

The follow-up review reproduced every tile from the live tables with the page's own formulas, so the arithmetic is faithful and the semantics are the problem. Of the 1,060 rows, 587 have no due date and 473 have one; every one of those 473 is in the past (latest 7 Sep 2026), which is why "Over Due" is 473 and "7 Days Due" is structurally zero. All 403 case/task rows (DRC-01, ASMT-10, LUT, enforcement...) were written with no due date although the case-folder JSON already stored holds 159 due dates and 35 hearing dates. "Last 24 Hours" = 64 is the six clients synced on 12 Sep; the newest issue date in that batch is 17 Jul. Of the 167 "open" DRC-03 cases, 131 are portal acknowledgements of accepted payments; of the 41 "open" refunds, 22 already ended in an RFD-06 / PMT-03 order. No row has a staff status, priority or owner set, which is consistent with the sync wiping them or with nobody having entered them; either way, no staff work currently survives a sync. No "PULL FAILED" rows exist today.

4 What is broken today (32 verified defects)

4.1 Critical and high defects (12)

Each item below was raised by at least one reviewer and survived two adversarial verification passes; file references point at the current `main`. Severity: ■ critical = wrong data or data loss in production; ■ high = a number or link a partner would rely on is wrong; ■ medium = incorrect in edge cases or hides errors; ■ low = polish. Effort is S (hours), M (days), L (a week or more). All of these are scheduled in Phase 0 except where the fix column names a later phase. The 20 medium and low defects are listed in Appendix D.

#	Severity	Defect and impact	Fix	Eff.
B1	■ Critical	Every Sync All wipes staff workflow state on <code>gst_notices</code> (delete-then-insert), so no alert can ever be based on Open/Closed/assignee/reply state <code>extension/background.js:102</code> Staff mark a notice Closed / High priority / assigned to X, the next Sync All silently erases it. Beyond the data loss, this makes any state-based alerting (due ladder that stops when Closed, assignment emails, status-change events, 'new notice' detection by INSERT trigger) impossible — an INSERT trigger would fire for all ...	1) ALTER TABLE <code>gst_notices</code> ADD COLUMN <code>origin</code> text NOT NULL DEFAULT 'portal' CHECK (origin IN ('portal','manual')), ADD COLUMN <code>portal_missing_at</code> timestampz; 2) CREATE UNIQUE INDEX <code>uq_gst_notices_portal_key</code> ON <code>gst_notices</code> (<code>client_id</code> , <code>source</code> , <code>reference_number</code>) WHERE <code>origin</code> = 'portal' AND <code>reference_number</code> IS NOT NULL; CREATE ...	M
B2	■ Critical	clients.gst_password stored in plaintext and readable with the public anon key under open RLS <code>extension/background.js:63</code> Portal credentials for ~92 GSTINs are one curl away for anyone who opens the site's JS; a leak lets a third party file returns, download every notice, or change the client's registered e-mail. This underpins the entire notice sync so it belongs in the litigation data model review.	Move credentials out of the anon-readable table: CREATE TABLE <code>client_portal_credentials</code> (<code>client_id</code> uuid PK REFERENCES <code>clients</code> ON DELETE CASCADE, <code>gst_user_id</code> text, <code>gst_password_enc</code> bytea NOT NULL, <code>updated_at</code> timestampz DEFAULT now()); ALTER TABLE <code>client_portal_credentials</code> ENABLE ROW LEVEL SECURITY; (no policies — same pattern ...	M

#	Severity	Defect and impact	Fix	Eff.
B3	■ Critical	A failed portal read replaces the client's entire notice history with one fake 'PULL FAILED' row <code>extension/content.js:2471</code> One expired session or CAPTCHA failure deletes years of captured notices and PDFs for that client and inserts a row with notice_type NULL, which classifyNoticeCategory (src/utlis/noticeCategoryClassifier.ts:47) buckets as 'Uncategorised' and NoticesDashboardPage counts as an Open notice. Data loss plus a phantom notice.	Stop writing sentinel rows; log failures only via logClientSync (client_sync_log). Then enforce in DDL: DELETE FROM gst_notices WHERE description LIKE 'PULL FAILED%'; DELETE FROM gst_refund_applications WHERE status LIKE 'PULL FAILED%'; DELETE FROM gst_drc03_filings WHERE status LIKE 'PULL FAILED%'; ALTER TABLE gst_notices ADD ...	S
B4	■ High	email_outbox schema physically blocks team alerts: kind CHECK, client_id NOT NULL, single to_email, template_key FK, no dedupe key <code>supabase/migrations/20260725120000_gst_reminders.sql:45</code> Any INSERT with kind='notice_alert' or a NULL client_id is rejected by Postgres today, so the whole notice-alerting layer cannot be built without a schema migration first; without a dedupe key a re-run of a ladder job double-emails the team.	Migration <code>`2026xxxx_email_outbox_notice_alerts.sql`:</code> <code>`ALTER TABLE email_outbox ALTER COLUMN client_id DROP NOT NULL; DROP CONSTRAINT email_outbox_kind_check; ADD CONSTRAINT email_outbox_kind_check CHECK (kind IN ...</code>	S
B5	■ High	Case-folder notices (DRC-01, ASMT-10, Enforcement, Rectification...) never get a due date, so they are invisible to Over Due / 7 Days Due / calendar <code>extension/content.js:2613</code> The highest-stakes litigation items in the firm (show-cause notices under s.73/74, scrutiny notices, enforcement cases) are exactly the ones with statutory reply deadlines, and the dashboard silently reports none of them as due or overdue. Staff relying on the '7 Days Due' tile will miss DRC-01 reply dates.	In the content.js task loop, after fetching folder items, compute due = latest `duedt` (raw.duedt or sdtls. <dtscn srscn remnd>.duedt) among NOTCE/INTIM items that have no later REPLY item, and write it to row.due_date (plus issue_date from the latest notice refdt rather than assignmentDt). Alternatively add a Postgres ...	M
B6	■ High	"Failed Logins" tile never counts failed logins; login errors are undetected and unlogged, and the job silently stalls <code>extension/content.js:484</code> A client whose password changed or whose account is locked produces 0 in 'Failed Logins' and blocks the whole Sync All queue for 20 minutes until the job is discarded (content.js:148). Meanwhile the tile can show non-zero for reasons that have nothing to do with logins (a Supabase write error). Staff cannot trust the number and ...	In handleLogin, after the Login click detect the portal's error element (.alert-danger / text matching invalid locked password) and write client_sync_log {action:'login', status:'failed', message} then advance(job); after 3 CAPTCHA retries do the same with message 'CAPTCHA not solved'. Rename the tile 'Sync failures' with a ...	M

#	Severity	Defect and impact	Fix	Eff.
B7	High	Any job older than 20 minutes is silently discarded, so a firm-wide Sync All cannot finish <code>extension/content.js:148</code> Staff believe they synced everyone; in reality only the first handful of clients were refreshed and the rest keep stale notices, with no failed row in <code>client_sync_log</code> for the skipped clients and no indication on the dashboard.	Change the guard to a last-activity timestamp (<code>job.lastProgressAt</code> updated in <code>setJob</code>) with e.g. a 10-minute idle cap, and reset it in <code>advance()</code> ; on expiry write <code>client_sync_log</code> failed rows for the current client and mark the run aborted in a <code>sync_runs</code> table so the dashboard can show 'Run aborted at client 7/92'.	S
B8	High	Extension bridge only injects on *.vercel.app, so the production domain always shows "Extension not detected"; message also wrong on non-Chrome/mobile <code>extension/manifest.json:32</code> On the firm's real URL, Sync All can never start from the dashboard (the click path <code>handleSyncAll</code> at line 223 returns 'browser extension not detected'), so staff fall back to the extension popup and lose the <code>notices_bundle</code> chaining and the <code>client_sync_log</code> rows. On phones/tablets the hint tells users to install something ...	Add the production origin (and any staging origin) to <code>manifest.json</code> <code>content_scripts</code> matches for <code>appbridge.js</code> and bump the version; in the app derive a three-state indicator: (1) not desktop Chrome/Edge → 'Sync All is available on desktop Chrome with the GST Keeper extension' and hide the button; (2) Chrome but no reply within ...	S
B9	High	GSTIN Wise Notice Count is capped at 1000 rows and double-counts refund/DRC-03 cases, so it disagrees with Notice Summary <code>src/pages/GstinWiseNoticeCountPage.tsx:63</code> Reproduction: open Report > GSTIN Wise Notice Count and compare its Total row with Report > Notice Summary's Total — they differ (about 60 rows dropped by the cap, ~20 refund/DRC-03 cases counted twice). A partner using this page to see which client carries the most open matters gets wrong per-client numbers, and the last ~60 ...	Use <code>fetchAllRows</code> for <code>gst_notices</code> (select <code>case_id</code> and <code>notice_type</code> too) and reuse the per-category dedupe: skip <code>notice_type</code> in ('Refunds','Voluntary Payment') in the notices loop, and merge dedicated refund/DRC-03 rows with those case rows by ARN/case_id per client exactly as <code>mergeIntoCategory</code> does — ideally by extracting that ...	M
B10	High	Over Due (473) includes closed and replied notices, ignores extended_due_date, and counts today-due items as overdue from 05:30 IST <code>src/pages/NoticesDashboardPage.tsx:257</code> 473 of 1060 'overdue' next to '7 Days Due 0' is the contradiction the principal noticed. The number is inflated by historical, already-handled notices and by same-day due items, so the firm cannot use it to drive urgency or to email the GST team about genuine lapses.	Define <code>effectiveDue = extended_due_date ?? due_date</code> and <code>actionable = !isClosed(staff_status) && !reply_date && !submission_date</code> . <code>overdue = actionable && effectiveDue < today</code> ; <code>dueSoon = actionable && today <= effectiveDue <= today+7</code> . Put this in one helper in <code>src/utils/noticeSummaryReport.ts</code> (e.g. <code>noticeDueState(row)</code>) and use it ...	S

#	Severity	Defect and impact	Fix	Eff.
B11	High	'Last 24 Hours' tile counts every notice of every client synced today, not new notices <code>src/pages/NoticesDashboardPage.tsx:254</code> Reproduction: run Sync All for one client with 40 old notices; 'Last 24 Hours' jumps by 40 although nothing new was served. The tile cannot distinguish 'newly received' from 'recently re-downloaded', so it cannot be used as the 'what arrived since yesterday' signal the label implies; its drill-down label ('pulled from the ...	After moving to upsert (finding 1), keep created_at as first-seen and compute last24Hours/last15Days from created_at (or a dedicated first_seen_at) rather than pulled_at; keep pulled_at as 'last verified on portal'. Rename the drill-down label accordingly.	S
B12	High	No natural/unique key on gst_notices, gst_refund_applications, gst_drc03_filings — duplicates already present <code>supabase/migrations/20260817180000_notices_refund_drc03.sql:27</code> Counts on the Notice Summary, Refund and DRC-03 pages depend on each page re-implementing the same ad-hoc dedupe; any page that forgets (GstinWiseNoticeCountPage, CompanyProfilePage) reports different totals for the same client. Without a key there is also nothing to upsert against, which forces the destructive ...	After a one-off cleanup (DELETE duplicates keeping min(created_at)): CREATE UNIQUE INDEX uq_gst_refund_arn ON gst_refund_applications (client_id, arn) WHERE arn IS NOT NULL; CREATE UNIQUE INDEX uq_gst_drc03_arn ON gst_drc03_filings (client_id, arn) WHERE arn IS NOT NULL; plus the partial unique indexes on gst_notices from the ...	S

Verification record

None of the first-pass "broken" claims was refuted: two verifiers per review lens traced each claim from the data source to the rendered value and tested it against realistic row shapes; several corrected a line number or sharpened the fix, and those corrections are reflected above.

4.2 Further defects from the follow-up reviews (22 critical and high; 9 more in Appendix D)

The completeness critic named six areas the first pass had not covered (live-data reconciliation, the trust boundary between app, extension and storage, deletion and retention, role-by-role permissions, institutional knowledge, and the refund / DRC-03 surfaces). These reviews ran after the adversarial pass, so their items were not independently refuted. The author hand-checked the most severe: the production-domain manifest gap, the bridge's missing origin check, the public bucket and its "anyone can delete" policies, the role restored from localStorage, and the unguarded Clients route are all confirmed in the current code. Rows below are numbered on from the table above.

#	Severity	Defect and impact	Fix	Eff.
B33	■ Critical	Extension app bridge accepts postMessage from any origin/frame and can drive portal logins, Sync All, GSTR-1 upload and GSTR-3B push <code>extension/appbridge.js:15</code> Any page on any Vercel subdomain (a stranger's free deployment) or any iframe embedded in the app can post <code>{__gstkPullSectionAllClients: {mode:'notices_bundle'}}</code> and the firm's browser will log every one of the 121 credentialed clients into the GST portal, or post <code>{__gstkPushGstr3b: {clientId, period_month, gstr3bJson}}</code> with ...	In appbridge.js reject unless <code>`e.source === window && e.origin === location.origin`</code> ; in manifest.json replace the wildcard with the exact production origin(s) (<code>`https://gst.vjdesai.com/*`</code> plus the named preview host); in background.js check <code>`sender.url`</code> origin against an allow-list before dispatching, and whitelist <code>`msg.fn`</code> to ...	S
B34	■ Critical	return-pdfs bucket is public with permanent URLs and guessable paths — every scanned notice/order PDF is fetchable by anyone, no auth <code>extension/background.js:242</code> Show-cause notices, demand orders, appeal papers, refund deficiency memos and registration certificates for 121 clients are on the open internet for anyone who has or can guess a link. The path is <code>clientId + the portal reference number</code> (printed on every notice), and <code>clientIds</code> are readable from the clients table with the anon ...	Set <code>`storage.buckets.public = false`</code> for return-pdfs (an UPDATE, not an INSERT ... ON CONFLICT DO NOTHING). Stop storing the <code>/object/public/</code> URL: store only the object path in <code>gst_notices.pdf_url / gst_case_folder_items.attachments[].url / gst_refund_applications documents / gst_drc03_filings.pdf_url</code> (a one-off migration can ...	L
B35	■ Critical	Extension app bridge is not registered for gst.vjdesai.com, so Sync All / Portal-open / Login are dead for every role on the live domain <code>extension/manifest.json:27</code> Regardless of what permission model is designed, no staff member on the production domain can trigger a notice sync, open a client on the portal from a notice row (NoticeWorkflowListView.tsx:241-248), or use Company List Sync/Fetch Company (CompanyListPage.tsx:225-243). The firm believes it is a permission/installation problem ...	Add <code>`"https://gst.vjdesai.com/*"`</code> to that <code>content_scripts `matches`</code> array (and drop the over-broad <code>`https://*.vercel.app/*`</code> wildcard, see the separate appbridge origin finding), bump manifest version, and document the domain in extension/README.md. Optionally have NoticesDashboardPage show the current origin in the 'Extension ...	S
B36	■ Critical	Production domain gst.vjdesai.com is not in the extension manifest — appbridge only loads on *.vercel.app, so Sync All is dead on the live site <code>extension/manifest.json:29</code> On the firm's real URL the dashboard's Sync All, the Company List Sync/Fetch Company buttons and every Reports Hub 'Pull' button believe the extension is absent, so staff must open the vercel.app preview URL to sync — a fact that lives in nobody's head or document.	Add <code>"https://gst.vjdesai.com/*"</code> to the appbridge <code>content_scripts matches</code> in extension/manifest.json (and record the list of allowed app origins in the README and in docs/NOTICES_LITIGATION_FLOW.md §Sync procedure); bump the manifest version so Company List's build= debug rows show which build staff are on.	S

#	Severity	Defect and impact	Fix	Eff.
B37	■ Critical	Session role is restored from the localStorage blob, not from user_roles — a client user can promote itself to superadmin without devtools tricks beyond editing one JSON key <code>src/contexts/AuthContext.tsx:204</code> A client who logs in normally (row exists in user_roles as 'client', so `roleData` is truthy), then edits `vjdesai_user` in Application > Local Storage to `{"role": "superadmin"}` and reloads, gets the full staff shell: Notices Dashboard, all clients' notices, the Credentials tab, Sync All, Edit Client, User Control. Because RLS ...	On bootstrap set `role: roleData.role` (server value) and drop the session if it differs from the stored one; better, have `get_user_snapshot` return role + permissions and treat localStorage only as the id. Long-term, sign the session (an RPC that returns a short-lived token verified by a `current_app_user()` SQL function) so ...	S
B38	■ Critical	/clients has no staff gate: a client-role login can open the Credentials tab and read all 121 GST portal passwords unmasked <code>src/pages/ClientsPage.tsx:66</code> The sidebar hides the link for clients, but typing /clients into the address bar is enough. A client user (or anyone who borrows a client's login) sees every other client's GST portal user id and password in plain text, with a one-click Excel/PDF export for managers (allClientsReports.ts:362-379 'Client Login Credentials'). ...	Add `if (!isStaffRole()) return` <Navigate to="/dashboard" replace />` to ClientsPage, EditClientPage and every notices page that lacks it (AddNoticeDialog/NoticeWorkflowListView have no gate), and a route-level `<StaffRoute>` wrapper in App.tsx so a new page can't forget. Introduce a `view_client_credentials` permission key in ...	S
B39	■ Critical	Client delete cascades through all litigation tables with no archive, no audit entry and no alternative offered <code>src/pages/CompanyListPage.tsx:193</code> One click by any employee holding delete_clients removes a taxpayer's complete SCN/order/appeal/refund/DRC-03 evidence trail and even the version and sync logs that would prove it existed. Appeal limitation periods and s.36 CGST retention (72 months, longer while proceedings are pending) make this a professional-liability ...	(a) Add `deleted_at` timestamp, `deleted_by` uuid, `delete_reason` text to clients; change the delete handlers to an UPDATE that sets these and hide soft-deleted clients in every list; (b) change the client FK on gst_notices, gst_case_folder_items, gst_refund_applications, gst_drc03_filings, client_sync_log, ...	L
B40	■ Critical	Dashboard Total/Open Notices tiles count 197 refund/DRC-03 case stubs as open notices <code>src/pages/NoticesDashboardPage.tsx:256</code> The headline KPI the principal reads first is inflated by acknowledgements of the firm's own voluntary payments; it can never fall below the stub count and hides the real open workload.	Exclude CASE_REFUND_TYPE/CASE_DRC03_TYPE rows from displayRows for the tiles (or route them through the same derived-state logic), and add separate 'Open refunds' / 'DRC-03 pending' tiles fed by the corrected merge.	S

#	Severity	Defect and impact	Fix	Eff.
B41	 Critical	DRC 03 'Open 167' counts 131 acknowledged payments as open <code>src/utils/noticeSummaryReport.ts:116</code> The Notice Summary tells the partner 167 DRC-03 matters are pending when 36 are; acknowledged voluntary payments (DRC-04 issued) are closed matters and inflate every open count, per-GSTIN count and any future email digest.	In mergeIntoCategory treat a case-only row as closed when its description matches /acknowledgement of acceptance drop proceedings/i (DRC-03) or its latest event is RFD-06/PMT-03 (refund), or better: derive closure from the latest <code>gst_case_folder_items</code> event per <code>case_id</code> ; keep <code>staff_status</code> as an override only.	S
B42	 High	extension/README.md is stale — describes the v1 filing-status/ledger extension and never mentions notices, refunds, DRC-03, Sync All or the app bridge <code>extension/README.md:13</code> The README is the only install/usage text a new staff member or a future Claude session gets; it actively misdescribes the extension (says 2B is not yet supported, says nothing about the notices module that is now its main job), so nobody learns how Sync All, the 20-minute expiry, or the PULL FAILED sentinel behave.	Rewrite extension/README.md: a current capability list (return pulls, 2B/2A downloads, notices_bundle Sync All, refund document harvest, DRC-03, GSTR-3A PDFs, Filing Status Portal button), the app-bridge detection handshake and the list of app origins in the manifest (add <code>gst.vjdesai.com</code>), job lifecycle (20-minute expiry, ...	S
B43	 High	Scanned notice/order PDFs are never deleted, leaving orphaned evidence in a public bucket while the row that points to them is gone <code>extension/background.js:242</code> After a client delete or a sync wipe the PDFs still exist but nothing in the app knows their path (<code>pdf_url</code> died with the row), so the evidence is effectively lost to the firm while still being world-readable by URL. Conversely, because the DELETE policy is open to public and the anon key ships inside the SPA bundle and the ...	(1) Record every upload in a 'litigation_documents' table (<code>client_id</code> , <code>table_name</code> , <code>row_id</code> , <code>path</code> , <code>sha256</code> , <code>uploaded_at</code> , <code>uploaded_by</code>) so documents are discoverable independent of the row; (2) never delete objects on row delete — instead the archive trigger from the history finding keeps <code>pdf_url</code> in <code>old_row</code> ; (3) replace the public ...	L
B44	 High	CONFIRMED capture gap: extension writes due_date:null for every case/task row although the portal folder JSON carries the due and hearing dates <code>extension/content.js:2613</code> The litigation cases (SCN, ASMT-10, appeal, audit) are exactly the rows with real reply deadlines and hearing dates, and they are the rows whose deadlines the dashboard can never show; a hearing tomorrow is invisible.	In the task-row loop, after fetching folders, derive <code>due_date</code> = latest of the due-ish keys above (parse <code>dd/mm/yyyy</code>) and add a 'hearing_date' column to <code>gst_notices</code> ; alternatively add a SQL view over <code>gst_case_folder_items</code> that extracts these keys so the dashboard can join without an extension release.	M

#	Severity	Defect and impact	Fix	Eff.
B45	High	Sync All bundle has never populated Refund/DRC-03 tables firm-wide; 32 clients have only amount-less stubs <code>extension/content.js:2316</code> The dashboard's DRC 03 row is 130/183 stubs and Refund 23/60 stubs with no amounts, statuses or PDFs because the dedicated pull ran for a handful of clients; the tooltip on the Sync All button (NoticesDashboardPage.tsx:425) promises all three sections.	Run a one-off firm-wide 'notices_bundle' sync and log each of refunds/drc03 as its own client_sync_log action with status, so the Company List 'Status' column and the Failed Logins tile reflect all three sections; surface 'last refund/DRC-03 pull' per client.	M
B46	High	sanctioned_amount is hard-coded null although RFD-05/06 orders with sanctioned figures are already captured <code>extension/content.js:2915</code> The report shows every refund as sanctioned Rs 0 and a 0 total, which is false and dangerous for a CA firm reconciling client refund receivables; the true figure is already in the database.	When handleRefundDocs (or a DB-side job) sees an RFD-05/RFD-06 order for an ARN, patch <code>gst_refund_applications.sanctioned_amount</code> (sum of <code>netsancamt.tot</code>), <code>rejected/inadmissible amount</code> and <code>order_date</code> ; show 'not yet pulled' instead of 0 when null.	M
B47	High	RFD-08 reply due dates are captured but dropped, so refund SCNs never reach 7 Days Due / Over Due / calendar <code>extension/content.js:2613</code> An RFD-08 has a 15-day statutory reply window (rule 92(3)); a missed reply means rejection and PMT-03 re-credit. Today the dashboard's due-date tiles, calendar and any reminder logic are blind to every refund deadline.	When the notices step captures a NOTAC item with <code>sdtls.duedate</code> , write it to the parent <code>gst_notices</code> row's <code>due_date</code> (and <code>issue_date</code> from <code>refdt</code>); add a deficiency-memo flag so RFD-03 cases show 'fresh application required' with the s.54(1) two-year limitation date.	M
B48	High	Production origin <code>gst.vjdesai.com</code> is not in the extension's <code>content_scripts</code> matches — Sync All / Portal buttons cannot work on the live site <code>extension/manifest.json:32</code> On the production domain the appbridge is never injected, so the <code>__gstkAppReady</code> ping (NoticesDashboardPage.tsx:214) is never answered: Sync All, the Company List Sync/Fetch buttons, the per-row Portal icon, GSTR-1 upload and GSTR-3B push are all dead for staff using the real URL. Only someone browsing the <code>*.vercel.app</code> preview ...	Add <code>'https://gst.vjdesai.com/*'</code> (exact) to the third <code>content_scripts</code> entry and to <code>host_permissions</code> if needed, remove the <code>*.vercel.app</code> wildcard, bump manifest version and re-load the unpacked extension on every staff machine. Document the domain in <code>extension/README.md</code> .	S
B49	High	DRC-03 drill-down shows Rs 0 for 130 of 183 rows while the same cases' payment detail (Rs 6.46 Cr) is already captured in case folders <code>src/pages/AllClientsDrc03Page.tsx:98</code> The firm's headline 'tax paid under protest / voluntarily' is understated roughly ten-fold on the only page that lists DRC-03s across clients, and the case-only rows label the cause of payment as 'Acknowledgement of acceptance'.	Backfill <code>gst_drc03_filings</code> from <code>gst_case_folder_items.raw_json.vp.py</code> sum (same parser as <code>parseDrc03Case</code>) during the notices pull, so every VP case has one dedicated row; drop the stub-synthesis branch in the page.	M

#	Severity	Defect and impact	Fix	Eff.
B50	High	Every staff role can read every client's GST portal password in plain text; only the export is restricted <code>src/pages/ClientsPage.tsx:57</code> Portal credentials are the single most sensitive asset in a GST litigation practice — with them anyone can file replies, withdraw appeals or change the authorised signatory. An article or a departing employee has them for all 121 clients by default, and the sidebar 'Clients' item is visible to every staff role ...	Add a `view_portal_credentials` key (AuthContext + UserControlPage). Hide the Credentials tab and GSTPortalLink unless it is granted; default showPasswords to false; keep export as superadmin/gst_manager. Move the 'log in as client' extension actions (ClientsPage.tsx:128, NoticeWorkflowListView.tsx:241) behind the same key so ...	M
B51	High	/add-client and /edit-client/:id have no permission guard — hiding the pencil is cosmetic, any staff can edit any company incl. its GST portal password by URL <code>src/pages/EditClientPage.tsx:51</code> An employee with no keys (an article) can type /edit-client/<id> — the id is visible in every /notices-company/<id> link they are allowed to open — and change the client's trade name, GSTIN, email, and GST portal password, which then breaks the firm's next portal sync for that client. The add_edit_clients key the principal ...	At the top of both pages, after hooks: `if (!canAddEditClients()) return <Navigate to="/clients" replace />` (mirror the pattern at UserControlPage.tsx:326-328). Longer term, move client writes behind a SECURITY DEFINER RPC that validates a staff session token so the anon key cannot bypass it.	S
B52	High	CONFIRMED: 'Open Notices' equals 'Total Notices' because status is staff-only and nobody has ever set it; portal evidence of closure is ignored <code>src/pages/NoticesDashboardPage.tsx:256</code> An Open count that can only ever fall when a staff member hand-closes each of 1060 rows is not a workload metric; the firm cannot see which matters actually need a reply.	Introduce a derived `effective_status` (pure function, tested): portal-closed when the case has an ORDRS/CLOSR folder item, the description begins 'Order'/'Registration Certificate', or the dedicated refund/DRC-03 status is terminal; staff_status overrides when set. Show 'Needs action' = effective open AND (due_date ≥ today OR ...	L
B53	High	CONFIRMED: 'Over Due 473' is every row that has any due_date; '7 Days Due 0' is structural, not a coincidence <code>src/pages/NoticesDashboardPage.tsx:257</code> The one tile that should drive the day's work ('what is due this week') is guaranteed to read 0, and the alarm tile is dominated by seven-year-old auto-generated GSTR-3A notices.	Compute overdue/due-soon only for effective-open rows, cap the look-back (e.g. due within last 90 days) or bucket by ageing (0-7 / 8-30 / 31-90 / 90+ days), and fix the case-row due_date capture (next finding) so future dates exist at all.	M
B54	High	Refund 'Open 41' includes 22 cases already closed by RFD-06/PMT-03 order <code>src/utils/noticeSummaryReport.ts:117</code> Refund follow-up lists show 41 pending claims when the portal has already disposed of most of them; staff chase closed matters and miss the real 8 'Refund Application filed' + 2 deficiency-memo cases.	Compute refund closure from the latest case event (ORDRS item / PMT-03 / RFD-01W) and treat 'Refund Sanctioned' as a distinct 'Sanctioned – awaiting disbursement' state rather than open; expose the derived state as a column on gst_refund_applications.	M

5 Look and feel: from counts to action

5.1 Design principles for the new page

- 1. Lead with action, not counts.** The first screen answers "what must the team do today" with real notices, owners and deadlines. Counts become context around that queue, never the whole page.
- 2. Every number must be defensible.** A tile is shown only if its definition is written down (in a tooltip and in this document) and its drill-down list contains exactly the rows it counted.
- 3. Money on the page.** Demand under dispute, pre-deposits paid and exposure by client are the numbers a partner manages by; they come from columns the schema already has.
- 4. One visual language.** Use the firm's existing tokens (navy primary, coral accent, success/warning/info/destructive) at the app's standard sizes: body 14px, captions 12px, KPI values 30px, 8-point spacing. No 10–11px text.
- 5. Two-week horizon.** Deadlines are shown as the next 14 days with counts per day and hearings marked; the month grid moves behind a toggle.
- 6. Health, not noise.** Sync status, failed logins and email delivery live in one compact "Sync & alerts" card with a last-synced time, instead of a KPI tile that mixes operations with legal metrics.
- 7. Client onboarding leaves.** Add / Import / Template belong on the Clients page; the dashboard keeps only a search box and a "Sync All" button.
- 8. Accessible by default.** Every icon button has a label, every clickable cell is a real link or button, focus rings are visible, and colour is never the only signal (chips carry text).

5.2 Target layout, section by section

Zone	What it shows	Data source / behaviour
Header	Title "Notices & Litigation"; sync line (last sync time, GSTIN count, failed logins in red); global search (GSTIN, trade name, ARN, case ID) with ⌘K; Add Notice; Send digest; Sync All (primary).	<code>client_sync_log</code> latest run; search hits <code>gst_notices</code> and <code>clients</code> . Sync All shows the extension/agent state as a chip, not a sentence.
Tabs + filters	Dashboard · Work Queue · Matters · Hearings · Clients · Reports. Filter chips: Type, FY, Owner, Officer/jurisdiction.	Active tab from the route (fixes the always-active "Dashboard" bug). Filters persist in the URL so a filtered view can be emailed.
Needs-attention strip (4 tiles)	Overdue & still open (with ₹ at risk and the oldest age) · Due in next 7 days (split by replies/appeals/hearings) · New since last sync (unassigned count) · Exposure under dispute (tax/interest/penalty split, pre-deposits, matters in appeal).	Each tile is one saved filter of the work queue. "Overdue" excludes closed and replied rows and honours <code>extended_due_date</code> ; "New" uses <code>first_seen_at</code> , not <code>pulled_at</code> .
Work queue (8 of 12 columns)	Notices sorted by statutory deadline: client, notice/matter, stage, due, days, demand, owner, action. Tabs Team / Mine / Unassigned / Hearings. Bulk toolbar: assign, priority, stage, email GST team, export.	Row click opens the matter drawer (§5.4). Footer names the stale-row problem and offers the closing sweep.

Zone	What it shows	Data source / behaviour
Ageing & exposure (4 columns)	Ageing buckets of open notices (not due, 1–7, 8–30, 31–90, 90+ days late); donut of demand by stage; top exposure by client; officer with most open matters.	recharts already in the bundle; colours from the semantic tokens; every bar/segment filters the queue.
Category summary	Horizontal stacked bars (open / replied / closed) for categories with data; empty categories collapse behind "+ n categories with no data".	Replaces the 18-row table with 8 permanent dash rows. Same <code>computeNoticeSummary</code> helper, corrected definitions.
Next 14 days	Two week strips with per-day chips: reply due, appeal limitation, hearing, issued. Toggle to month.	Computed and overriding due dates, hearings from <code>matter_hearings</code> . Day click opens the queue filtered to that date.
Sync & alerts	Last Sync All, clients synced / with credentials, failed logins (linked to fix), new/changed rows, emails sent today, extension version; a short activity feed; next unattended sync time.	<code>client_sync_log</code> (with the new <code>run_id</code> and row counts), <code>email_outbox</code> , <code>matter_events</code> .

5.3 Visual system

- **Type:** Poppins 600/700 for titles and KPI values, Inter for everything else. Sizes: page title 22px, card title 14px, body 13–14px, captions 12px, KPI 30px tabular.
- **Colour semantics (app-wide rule kept):** destructive = overdue / critical, warning = due soon / open, info = new / awaiting, success = closed / healthy, navy = neutral primary. Chips always carry text ("–2", "High", "Awaiting client data").
- **Cards:** 12px radius, 1px border, no drop shadows; a 4px coloured left rail on the attention tiles only.
- **Density:** table rows 40px; 8-point spacing; the page fills the width at 1600–1920px and stacks to one column below 1024px; on a phone the work queue becomes a card list.
- **Empty and error states:** every card has a real empty state ("No hearings in the next 14 days") and an error state with a retry, using the shared `TableEmptyState`.

5.4 The matter drawer (the page behind every row)

Clicking any notice opens a right-hand drawer (or a full page on small screens) that holds the whole matter: title, badges (priority, stage, owner, reviewer, section), a stage stepper with dates and the next statutory step, four fact tiles (demand breakdown, paid / pre-deposit, statutory deadline with its basis, client-data status), a warning banner when a deadline has passed, and tabs for Activity, Reply workspace, Documents, Demand & payments, Emails and Notes. The right column shows the client card, documents (portal PDFs and uploads together), key dates and every email sent on the matter. Every action available in bulk on the queue is available here on one item.

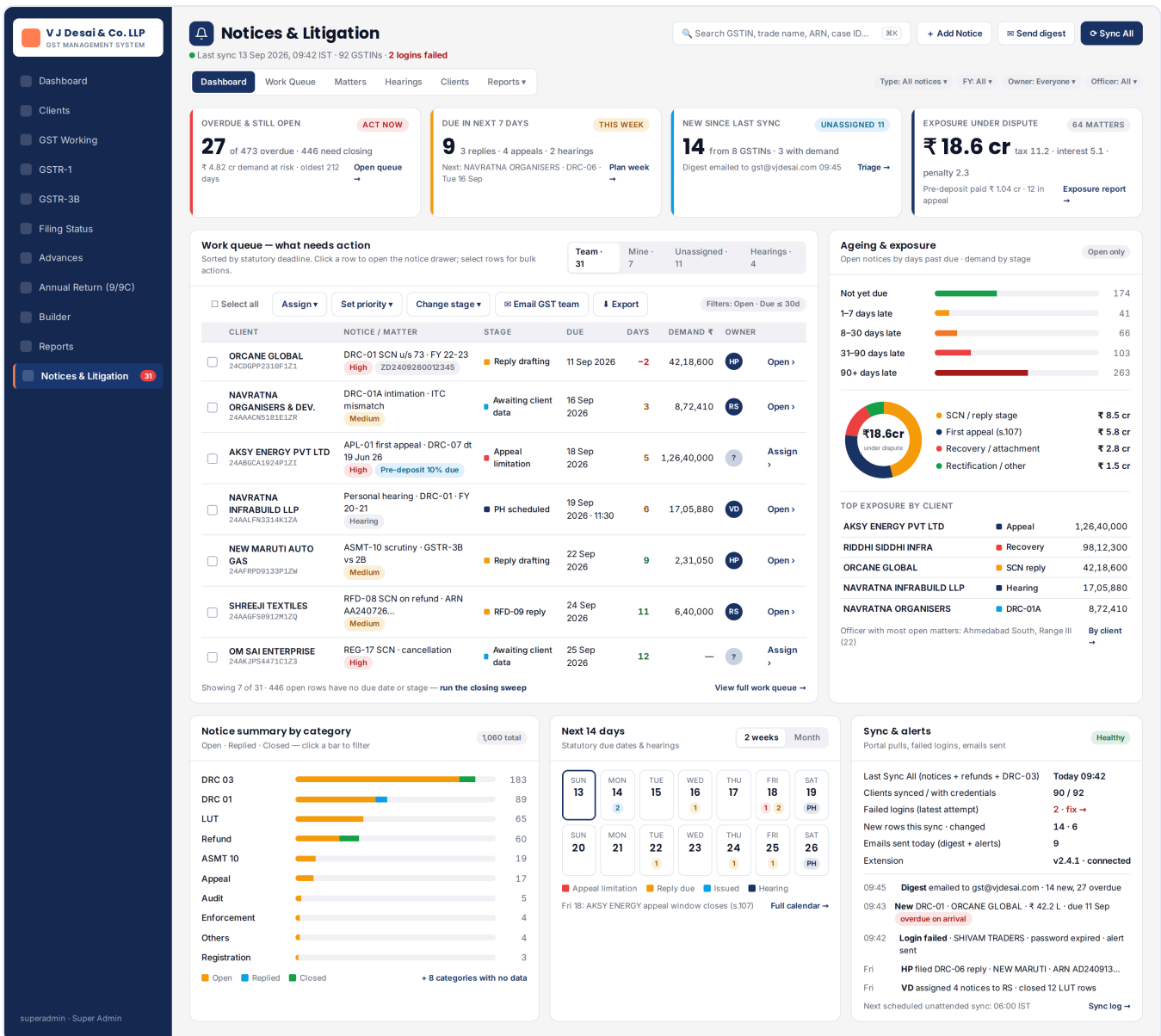
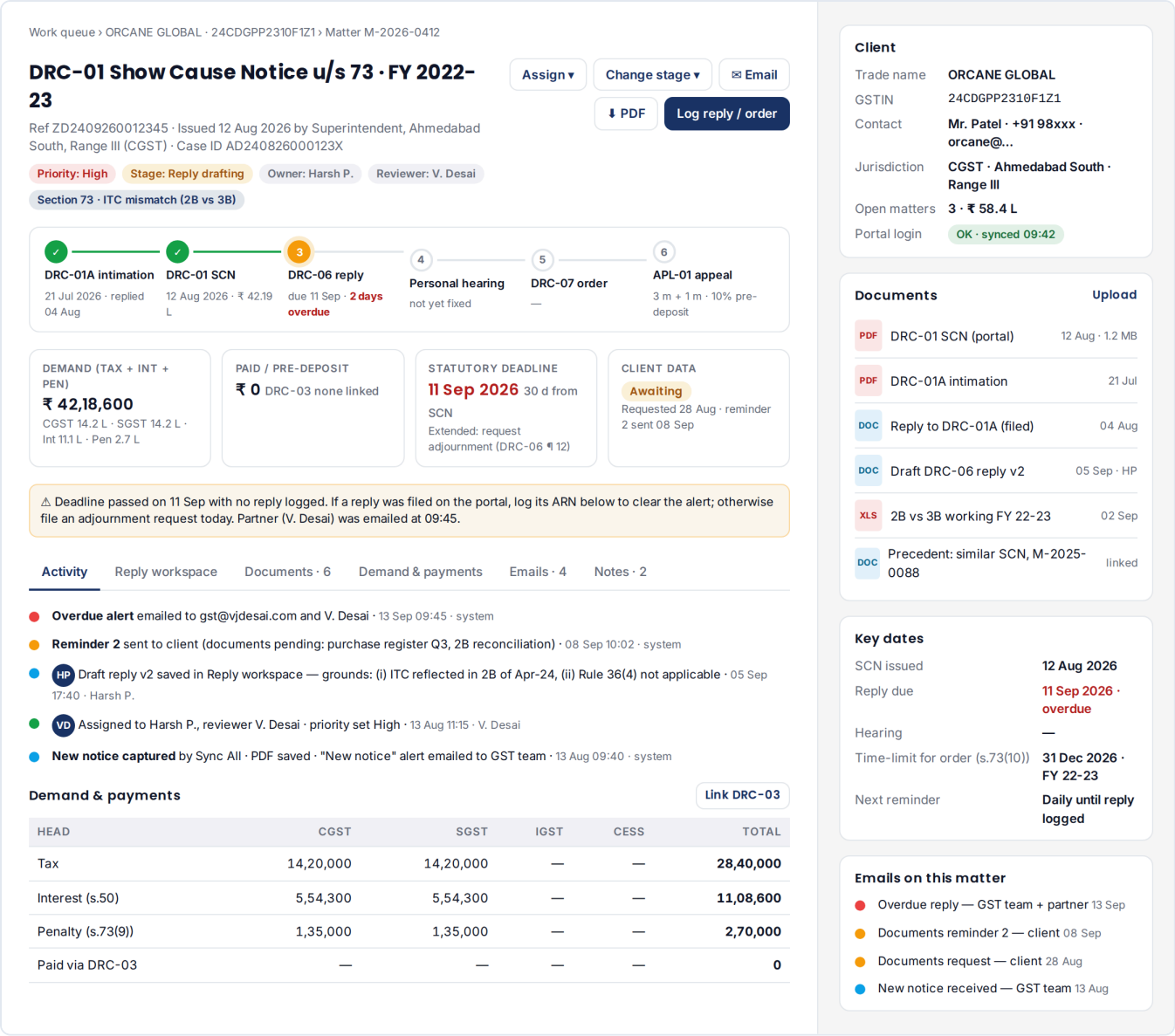


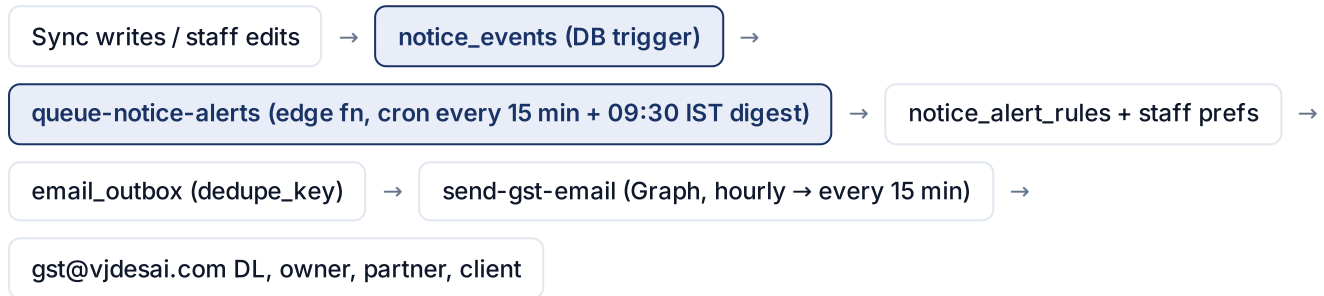
Figure 2. Redesigned Notices & Litigation dashboard at 1600 px. Needs-attention strip, work queue with bulk actions, ageing and exposure, category bars, 14-day deadline strip, sync and alerts. Sample data.



6 Emailing the GST team about every important matter

6.1 Principle: every important matter reaches the GST team's inbox without anyone opening the app

The firm already owns a working delivery pipeline: `email_outbox` rows are sent from `gst@vjdesai.com` through Microsoft Graph by the `send-gst-email` edge function, on an hourly `pg_cron` schedule, with staff-editable templates. Notice alerting reuses that pipeline end to end and adds only what is missing: an **event source** (what happened to a notice), a **rules engine** (who should hear about it, when) and **digest composition**. Nothing new sends email directly.



6.2 Event → recipient → template matrix

#	Event	Rule / schedule (IST)	Recipients	Template key	Content highlights
E1	New notice captured by sync (first_seen_at = this run)	Immediately after the client's sync completes; batched per client per run	GST team DL; auto-owner if a client→staff mapping exists	<code>notice_new</code>	Client, type, ref, issue/due dates, demand if any, portal PDF link, "Triage" deep link. High-priority types flagged in the subject.
E2	Daily digest	09:30 every day, after the 06:00 unattended sync; skipped only if nothing changed and nothing is due within 7 days	GST team DL; partner cc	<code>notice_digest_daily</code>	Sections: act today (overdue with demand), this week, new today, sync failures, unassigned. Mock in §5.
E3	Reply due ladder	T-7, T-3, T-1, T0 at 08:30; then daily while overdue for 5 days, weekly thereafter	Owner; GST team DL from T-1; partner from T0	<code>notice_due_T7 / T3 / T1 / overdue</code>	Days left, statutory basis, what is still missing (client data, draft, review), one-click "Log reply ARN".
E4	Hearing reminder	T-7, T-1 and 07:30 on the day	Owner, partner, counsel (if set), optional client	<code>hearing_reminder</code>	Venue/mode, officer, documents to carry (from <code>matter_documents</code>), previous hearing outcome.

#	Event	Rule / schedule (IST)	Recipients	Template key	Content highlights
E5	Appeal limitation on an order with no appeal filed	T-30, T-14, T-7, T-1 before the s.107 / s.112 date	Owner + partner (always)	limitation_alert	Order date, computed last date, condonation window, pre-deposit amount computed at 10%.
E6	Recovery / detention / registration cancellation captured (DRC-13/16/22, MOV-07, REG-17)	Immediately; priority subject prefix	Partner + GST team DL	notice_critical	Marked "critical" in subject; the digest repeats it until acknowledged in-app.
E7	Assigned / reassigned	Immediately	New owner (and previous owner on reassignment)	assignment	Matter summary, due date, reviewer, link.
E8	Stage change / closure	Immediately for closure and partner review; others in digest only	Owner, reviewer, partner on closure	stage_change	Old → new stage, who, note, order/ARN captured.
E9	Sync failed / login failed	Immediately after a failed attempt; repeated in digest until fixed	Admin (superadmin), GST team DL	sync_failed	Client, reason (password expired, CAPTCHA, portal down), last good sync, "Fix credentials" link.
E10	Weekly partner MIS	Monday 08:30	Partners, GST manager	mis_weekly	Exposure by stage/client/officer, ageing, opened vs closed, per-staff load, hearings next 2 weeks, PDF attachment.
E11	Unassigned for 24 h	Daily 10:00 while any exist	GST manager	unassigned_reminder	List with suggested owner (last person who handled that client).
E12	Client: notice received, documents needed	Staff-triggered from the drawer; reminders reuse the existing client_reminder_settings ladder	Client contact(s) from the client master; cc GST team	client_notice_docs, client_notice_docs_2	Plain-language summary, document checklist, due date for the firm (3 days before the statutory date), reply-by-email instruction.
E13	Client: status update / closure (optional per client)	On filing of reply and on closure	Client contact(s)	client_notice_update	What was filed, ARN, next expected step.

6.3 Design rules for the alerting layer

- **Idempotent.** Every alert carries a `dedupe_key` (rule + matter/notice + offset + date). Re-running the cron or re-syncing never double-sends, which the current outbox cannot guarantee for anything except filing confirmations.
- **Team address first.** Recipients are a distribution list (`gst@vjdesai.com` or a dedicated `gst-notices@`) plus roles, not individual addresses hard-coded in functions. Per-person preferences (`staff_notification_prefs`) can only reduce, never remove, the team copy.
- **One shell, many templates.** The existing corporate email shell (`_shared/email.ts`) gains a "notice" variant with a details box (client, GSTIN, type, ref, due date, demand) and an optional critical pill; templates stay staff-editable on the Reminders page, grouped under a new "Notice alerts" tab.
- **Attachments and deep links.** Graph supports attachments; the notice PDF (already in Storage as `pdf_url`) is attached to E1 and E6 when under 4 MB, otherwise linked. Every email deep-links to the matter drawer.
- **Quiet hours and batching.** Immediate events between 21:00 and 07:30 IST are held and folded into the 09:30 digest, except E6 (critical).
- **Auditable.** Every send is a `matter_events` row, so the drawer's "Emails on this matter" panel and the client's communication log come for free.
- **Fallback channel later.** The event log is channel-agnostic; WhatsApp (via a provider such as Gupshup or Interakt) or in-app bell notifications can subscribe to the same rules in a later phase without touching the rules engine.

Daily GST Notices Digest

To: gst@vjdesai.com
Cc: V. Desai (partner)

V. J. Desai & Co. LLP · GST Team · Saturday 13 Sep 2026, 09:45 IST

Sync All completed 09:42 · 90 of 92 GSTINs pulled · 2 logins failed (SHIVAM TRADERS, KHODIYAR STEEL) · 14 new notices · 6 changed

27

OVERDUE · OPEN

9

DUE IN 7 DAYS

14

NEW TODAY

2

HEARINGS THIS WEEK

₹18.6 cr

UNDER DISPUTE

ACT TODAY

Overdue replies with a live demand

ORCANE GLOBAL

24CDGPP2310F1Z1 · DRC-01 u/s 73 · FY 22-23 · Owner HP

₹ 42,18,600

due 11 Sep · 2 d late

Open >

RIDDHI SIDDHI INFRA

24AAECR1123B1Z7 · DRC-13 bank attachment · Owner —

₹ 98,12,300

recovery initiated 09 Sep

Assign >

KHODIYAR STEEL

24AABFK2011P1ZS · ASMT-10 · Owner RS

₹ 3,10,400

due 10 Sep · 3 d late

Open >

THIS WEEK

Statutory deadlines & hearings

AKSY ENERGY PVT LTD

APL-01 appeal window closes (s.107) · 10% pre-deposit ₹ 12.64 L

Fri 18 Sep

5 days

Open >

NAVRATNA INFRABUILD LLP

Personal hearing · DRC-01 FY 20-21 · Ahmedabad South R-III

Sat 19 Sep 11:30

6 days

Brief >

NAVRATNA ORGANISERS & DEV.

DRC-01A reply · awaiting client data (2 reminders sent)

Tue 16 Sep

3 days

Open >

NEW

Captured in today's sync (14) — 11 unassigned

ORCANE GLOBAL

DRC-07 order · FY 21-22 · ₹ 6,42,000 · appeal by 12 Dec 2026

UNASSIGNED

Triage >

OM SAI ENTERPRISE

REG-17 SCN · registration cancellation · reply 7 working days

UNASSIGNED

Triage >

SHREEJI TEXTILES

RFD-08 SCN on refund ARN AA2407... · reply 15 days

RS

Open >

+ 11 more (LUT acknowledgements 6, DRC-03 acknowledgements 4, ASMT-10 1) — [see all new >](#)

SYNC

Portal logins failed — credentials need attention

SHIVAM TRADERS

password expired on portal · last successful sync 28 Aug

Fix credentials >

KHODIYAR STEEL

invalid CAPTCHA 3× · retry scheduled 06:00

Retry now >

Open work queue

Weekly partner summary

You receive this because you are on the GST team distribution list. Per-notice alerts (new notice, overdue, hearing T-3) are sent separately as they happen. Manage recipients and quiet hours in [Settings >](#) Notice alerts. Sent by GST Keeper on behalf of V. J. Desai & Co. LLP.

Figure 4. The daily digest (event E2) as it would arrive at gst@vjdesai.com at 09:30 IST: act-today items with demand, this week's deadlines and hearings, new captures, failed logins. Sample data.

7 The litigation module: lifecycles, stages and pages

7.0 Module map

Page / surface	Purpose	Builds on
Dashboard	Needs-attention strip, work queue, ageing and exposure, category bars, 14-day strip, sync and alerts (figure 2).	Replaces <code>NoticesDashboardPage</code> ; keeps its route.
Work Queue	Full firm-wide list with saved filters (overdue, due 7d, unassigned, high priority, new, hearings), bulk actions, export.	Evolves <code>AllClientsNoticesPage</code> and <code>NoticeWorkflowListView</code> .
Matters	List and board (by stage) of litigation matters; create from any notice; filters by lifecycle, client, owner, officer, FY.	New page on <code>litigation_matters</code> .
Matter drawer / page	Everything about one dispute (figure 3): stepper, deadlines, demand and payments, hearings, documents, reply workspace, review, emails, notes.	Replaces the edit dialog; absorbs <code>AdditionalNoticeFolderPage</code> .
Hearings	Calendar and list of hearings and statutory dates; ICS export; adjournment logging; documents-to-carry checklist.	New page on <code>matter_hearings</code> .
Client profile > Notices	Per-client view: open matters, exposure, sync health, documents requested, one-click client status report.	Evolves <code>CompanyProfilePage</code> .
Reports	Notice summary and GSTIN-wise count (corrected), exposure by client/stage/officer/section, ageing, outcomes and time-to-close, per-staff load, client litigation status PDF.	Existing report kit and letterhead PDFs.
Settings > Notice alerts	Rules (events, offsets, recipients, escalation, quiet hours), templates, distribution list, per-person preferences.	Extends <code>RemindersPage</code> .
Client portal > Notices & litigation	Read-only stage, next date, documents requested, filed replies.	New tab in the client portal.
Permissions	<code>manage_notices</code> (edit, assign, stage), <code>close_matters</code> , <code>partner_review</code> , <code>view_exposure</code> ; superadmin and <code>gst_manager</code> implicit as today.	<code>user_permissions</code> keys and the <code>can*()</code> helpers.

7.1 The lifecycles the module must model

Each row in `gst_notices` today is a flat portal record. A litigation ERP treats a **matter** as the unit of work: one dispute that moves through statutory stages, each with its own form, deadline, document and owner. The table below is the reference the stage engine, the deadline calculator and the alert rules are built from. Statutory periods are stated as the app should default them; every deadline stays staff-editable because notices routinely specify a different date.

Lifecycle	Stages and forms (CGST Act / Rules)	Default deadline the app computes	Alert / action it should drive
Scrutiny	ASMT-10 notice → ASMT-11 reply → ASMT-12 acceptance, or escalation to audit / s.73–74	Reply within 30 days of ASMT-10 (or the date stated in the notice)	Reply ladder T-7/T-3/T-1; auto-request GSTR-3B vs 2B working from the 2B Reconciliation module

Lifecycle	Stages and forms (CGST Act / Rules)	Default deadline the app computes	Alert / action it should drive
ITC / liability mismatch	DRC-01B (Rule 88C, GSTR-1 vs 3B) and DRC-01C (Rule 88D, 2B vs 3B) intimations → Part B reply or payment via DRC-03	7 days from intimation	Same-day alert; block-filing warning surfaced on Filing Status for that GSTIN
Audit	ADT-01 (min. 15 working days' notice) → audit (3 months, extendable 6) → ADT-02 findings within 30 days of conclusion	Document list due date from ADT-01; ADT-02 expected date	Document checklist task; hearing/visit calendar entries
Demand (s.73 / s.74 / s.74A)	DRC-01A intimation (Rule 142(1A)) → DRC-01 SCN + DRC-02 statement → DRC-06 reply → personal hearing (s.75(4)) → DRC-07 order → DRC-08 rectification (s.161)	DRC-01A Part B reply as stated; DRC-06 reply 30 days from SCN (notice date governs); reduced-penalty payment window 30 days (s.73/74) or 60 days (s.74A); order time-limit 3 yrs / 5 yrs from annual-return due date, or 12 months from notice under s.74A	Reply ladder; hearing reminders; demand breakdown (tax/interest/penalty per Act); DRC-03 linkage; on DRC-07, auto-create the appeal-limitation clock
First appeal (s.107)	APL-01 with 10% pre-deposit of disputed tax (25% of penalty for s.129 orders) → APL-02 acknowledgement → hearing → APL-04 summary of order	3 months from communication of order, condonable by 1 month; pre-deposit before filing	Limitation alerts T-30/T-14/T-7/T-1 to owner and partner; pre-deposit amount computed and tracked as a payment
Tribunal (s.112, GSTAT)	APL-05 with additional 10% pre-deposit → GSTAT hearing → order; onward writ/appeal to High Court / Supreme Court	3 months from APL-04 (condonable 3 months); check the special window notified for legacy orders before relying on the default	Limitation alerts; counsel assignment; hearing calendar; cost tracking
Rectification (s.161)	Application for rectification of an order → DRC-08 rectified order	Application within 3 months of the order; authority may rectify within 6 months	Optional clock when an order shows an arithmetical/clerical error
Recovery (s.78–s.83)	DRC-09 (deduction), DRC-13 (garnishee to bank/third party), DRC-16 (attachment/sale), DRC-22 (provisional attachment, s.83, valid 1 year), DRC-20/21 instalments	Recovery may start 3 months after service of order (earlier if revenue interest is recorded)	Critical alert to partner within minutes of capture; stay-application task; show on the client's profile as a red banner
Refund (s.54)	RFD-01 (within 2 years of relevant date) → RFD-02 ack (15 days) / RFD-03 deficiency memo → RFD-04 provisional 90% (7 days) → RFD-08 SCN → RFD-09 reply → RFD-06 sanction/rejection (60 days) → RFD-05 payment / PMT-03 re-credit	RFD-09 reply within 15 days of RFD-08; deficiency memo requires a fresh RFD-01 (limitation continues to run)	Reply ladder on RFD-08; ageing of pending refunds beyond 60 days; interest (s.56) tracker
Voluntary payment	DRC-03 (and DRC-03A to adjust against a demand) → DRC-04 acknowledgement	None; link to the matter it settles	Reduce "demand outstanding" on the linked matter; close the matter when fully paid

Lifecycle	Stages and forms (CGST Act / Rules)	Default deadline the app computes	Alert / action it should drive
Registration	REG-17 SCN for cancellation → REG-18 reply → REG-19 cancellation / REG-20 drop; REG-21 revocation application → REG-23 SCN → REG-24 reply → REG-22 revocation order	REG-18: 7 working days; REG-21: 90 days from cancellation order (extendable); REG-24: 7 working days	Same-day alert (short clock); registration flag mirrored on the Clients page
Non-filer	GSTR-3A notice to return defaulter → file within 15 days, else best-judgement assessment (s.62, ASMT-13), withdrawn if return filed within 60 days	15 days from GSTR-3A	Route to the Filing Status reminder ladder that already exists
E-way bill / detention	MOV-01 statement → MOV-02 inspection → MOV-04 physical verification → MOV-06 detention → MOV-07 notice → MOV-09 penalty order (s.129) → MOV-11 confiscation (s.130)	Order within 7 days of MOV-07; appeal with 25% penalty pre-deposit	Critical, same-day alert (goods are detained); hearing reminder
Summons / enforcement	Summons under s.70 (appearance), search and seizure (s.67), DRC-01 that follows	Appearance date from the summons	Hearing-type reminder T-3/T-1 to partner and client
Amnesty (s.128A)	SPL-01/SPL-02 applications → SPL-05 order (or SPL-07 rejection → appeal)	Scheme windows have largely closed; keep the category for history and any reopened window	None by default; category remains reportable

Position to confirm with the partner

Default periods above follow the Act and Rules as generally applied; the notice text always governs and officers routinely grant shorter or longer windows. The engine therefore stores a *computed* deadline and an *overriding* deadline separately, and every alert states which one it used. Treat the 3-month appeal default and pre-deposit percentages as configuration held in a `litigation_rules` table, not as constants in code, so a Finance Act change is a data edit.

7.2 Stage model

Stages are a single ordered enum per lifecycle so that ageing, MIS and alerts can compare like with like. A matter moves forward through `matter_stage_history`; it never edits its stage in place without a history row.



- **Captured → Triage** is automatic on sync. A notice that arrives with a demand or a short clock (DRC-01B/01C, REG-17, MOV-07, DRC-13/16/22) is auto-tagged High priority.

- **Closed** requires a reason and, where relevant, the order number or DRC-03 ARN. LUT acknowledgements and DRC-03 acknowledgements are auto-closed by rule, which is what removes the 446 stale "open" rows the dashboard shows today.
- **Reopen** is allowed (an order can be followed by an appeal) and is logged.

8 Data model

8.1 Target schema (additive; existing pages keep working)

The model keeps `gst_notices`, `gst_refund_applications` and `gst_drc03_filings` exactly as the portal sync writes them and adds a **matter** layer above them. Nothing below removes a column the current pages read. Orange headers are new tables, blue headers are existing tables that gain columns.

litigation_matters

```
id uuid pk
client_id → clients
matter_no text unique (M-2026-0412)
lifecycle text
(demand|appeal|refund|...)
title, section_of_law,
financial_years text[]
authority (CGST|SGST),
officer, jurisdiction
stage text, status text,
priority text
owner_user_id,
reviewer_user_id → app users
demand_tax, demand_interest,
demand_penalty, demand_cess
numeric
paid_total,
pre_deposit_total numeric
computed_due_date,
override_due_date date
limitation_date date,
hearing_at timestamptz
next_action text, closed_at,
closed_reason
created_at, updated_at
```

gst_notices (+ columns)

```
matter_id →
litigation_matters
(nullable)
portal_key text (ref no ||
case id || hash)
first_seen_at, last_seen_at
timestamptz
portal_hash text (change
detection)
is_new bool, deleted_at
timestamptz (soft delete)
assign_to_user_id uuid
(replaces free text)
computed_due_date date
UNIQUE (client_id, source,
portal_key)
```

matter_stage_history

```
id, matter_id
from_stage, to_stage text
changed_by uuid, changed_at
note text
```

matter_events (activity log)

```
id, matter_id, notice_id
event_type text
(captured|assigned|stage|reply_logged|order|email_sent|note|document)
actor_user_id uuid (null = system)
payload jsonb, created_at
INDEX (matter_id, created_at desc)
```

matter_documents

```
id, matter_id, notice_id
kind
(notice|reply|order|annexure|working|hearing_note|precedent)
title, storage_path (private bucket, signed URLs), mime,
size
source (portal|upload)
uploaded_by, created_at
```

matter_hearings

```
id, matter_id
scheduled_at timestamptz,
mode (physical|video)
venue, officer, attended_by
uuid[]
outcome text, adjourned
bool, next_date
created_at
```

matter_payments

id, matter_id
 kind
 (pre_deposit|voluntary|recovery|refund_received)
 drc03_arn →
 gst_drc03_filings.arn
 tax, interest, penalty, cess
 numeric
 paid_on date, remarks

notice_alert_rules

key text pk
 (new_notice|due_ladder|hearing|limitation|failed_login|digest|weekly_mis|unassigned)
 enabled bool, offsets_days
 int[]
 recipients text[] (DL + roles)
 escalate_to text[],
 template_key
 quiet_hours int4range,
 updated_by

notice_alert_log

id, rule_key, matter_id,
 notice_id
 offset_days int, fired_on
 date
 outbox_id → email_outbox
 UNIQUE (rule_key,
 coalesce(matter_id,notice_id),
 offset_days, fired_on)

email_outbox (+ columns)

kind CHECK +=
 'notice_alert','digest','hearing','assignment','mis'
 client_id → NULLABLE
 (digests have no client)
 to_emails text[], cc_emails
 text[], attachments jsonb
 matter_id, notice_id uuid
 dedupe_key text UNIQUE
 priority smallint

client_sync_log (+ columns)

rows_new, rows_changed,
 rows_removed int
 duration_ms int,
 extension_version text
 run_id uuid (groups one Sync
 All)

litigation_rules

key text pk
 (reply_days.drc01=30,
 appeal_months=3,
 predeposit_pct.s107=10 ...)
 value numeric,
 effective_from date
 note text

staff_notification_prefs

user_id pk, email text
 digest bool, per_event jsonb
 quiet_hours, updated_at

8.2 Migration sketch (Phase 0 and Phase 3)

```
-- Phase 0: stop the data loss and make counts trustworthy
ALTER TABLE gst_notices
  ADD COLUMN portal_key text,
  ADD COLUMN first_seen_at timestamptz NOT NULL DEFAULT now(),
  ADD COLUMN last_seen_at timestamptz NOT NULL DEFAULT now(),
  ADD COLUMN portal_hash text,
  ADD COLUMN deleted_at timestamptz,
  ADD COLUMN computed_due_date date;
UPDATE gst_notices SET portal_key = coalesce(reference_number, case_id,
md5(coalesce(description,'')||coalesce(issue_date::text,'')));
CREATE UNIQUE INDEX uq_gst_notices_portal_key ON gst_notices (client_id, source, portal_key)
WHERE deleted_at IS NULL;
-- extension: POST gst_notices?on_conflict=client_id,source,portal_key Prefer:
resolution=merge-duplicates
-- and PATCH deleted_at for keys no longer returned – never DELETE the client's rows.

-- Phase 3: matter layer (all additive; a minimal matter_deadlines table arrives in Phase 1)
CREATE TABLE litigation_matters (...); -- see 8.1
ALTER TABLE gst_notices ADD COLUMN matter_id uuid REFERENCES litigation_matters(id);
CREATE TABLE matter_stage_history (...); CREATE TABLE matter_events (...);
CREATE TABLE matter_documents (...); CREATE TABLE matter_hearings (...);
```

```
CREATE TABLE matter_payments (...);    CREATE TABLE litigation_rules (...);  
-- RLS on every new table: FOR ALL TO public USING (true) WITH CHECK (true) (project rule)
```

Back-fill plan

A one-off script groups existing `gst_notices` rows into matters by `case_id` (the portal's ARN for case-linked rows) and, for rows without one, by `(client_id, financial_year, category)`. Staff then confirm groupings from a "Suggested matters" review page rather than editing 1,060 rows by hand. LUT and DRC-03 acknowledgements are auto-closed in the same pass.

9 Portal sync: making the data fresh and safe

9.1 What the sync does today

"Sync All" on the dashboard posts a message to the GST Keeper Chrome extension, which logs in to the portal for each client with saved credentials, asks a human to type one CAPTCHA per client, reads Notices & Orders, then Refunds and DRC-03 for the same session, and writes the results to Supabase. Three properties of this write path shape everything on the dashboard:

- **Delete-then-insert per client** (`replaceNotices` , `extension/background.js:102–105`): every sync deletes the client's `gst_notices` rows and inserts what the portal returned. Staff-entered status, priority, owner, reply and order details are lost each time, and `pulled_at` is reset for every row, so "Last 24 Hours" counts re-pulled rows, not new notices.
- **Failure replaces history** (`extension/content.js:2471, 2744`): a pull that fails writes a single diagnostic row with the description "PULL FAILED: ..." in place of all the client's notices. That row is then counted as a notice by every tile and report.
- **Not wired to the production domain** (`extension/manifest.json:32`): the app-side bridge script is injected only on `*.vercel.app` hosts. On `gst.vjdesai.com` the page never receives the extension's ready ping, so the dashboard shows "Extension not detected" (as in the screenshot) and Sync All cannot start there at all. Staff must be using the Vercel preview URL to sync. Adding the production host to the manifest is a one-line fix and is in Phase 0.
- **Fragile for a firm-wide run**: the extension drops any job older than 20 minutes (`content.js:148`), failures inside the refunds and DRC-03 steps are not logged, and portal passwords are copied in plain text into `chrome.storage.local` for every queued client (`background.js:410`).
- **Manual and browser-bound**: the sync exists only when a staff member is in Chrome with the extension enabled; there is no schedule, no retry, no record of how many rows were new, changed or removed, and the Windows agent runner (`agent/`) has no notices handler yet.

Legal posture stays as documented in `agent/README.md`

Automated access to the GST portal with stored client credentials is against GSTN's terms and carries IT Act exposure. This roadmap does not add CAPTCHA auto-solving. Unattended runs rely on the agent's existing session reuse (`storageState`), with a human entering one CAPTCHA per client session when it expires, written client authorisation on file, and the agent machine kept office-local. The partner should confirm this posture before Phase 4 is scheduled.

9.2 Target sync architecture



Change	Detail	Why
Upsert instead of replace	POST <code>gst_notices?</code> <code>on_conflict=client_id,source,portal_key</code> with Prefer: <code>resolution=merge-duplicates</code>, sending only portal-owned columns (reference, type, description, dates, issued_by, case_id, pdf_url, portal_hash, last_seen_at). Rows whose key is not returned this run get <code>deleted_at = now()</code> (soft delete). The same pattern already exists for <code>filing_status</code> in the extension.	Ends the data loss; keeps staff workflow columns; lets "new" and "changed" be computed from <code>first_seen_at</code> and <code>portal_hash</code> .
Failure never writes notices	A failed pull writes only a <code>client_sync_log</code> row with status <code>failed</code> , a reason class (login, captcha, portal_down, parse, db) and the step reached; existing notices are untouched.	Removes "PULL FAILED" rows from every count; gives the failed-logins tile a real definition.
Run summary	Each Sync All gets a <code>run_id</code> ; per client the log records rows_new, rows_changed, rows_removed, duration_ms, extension_version.	Feeds the Sync & alerts card, the digest header and the E9 alert.
Case rows get a due date	The case/task rows (DRC-01, ASMT-10, LUT, Enforcement, Rectification) are written with <code>due_date: null</code> today. The extension captures the portal's due date where the case JSON carries one; otherwise the app computes <code>computed_due_date</code> from the lifecycle rules once the row is attached to a matter.	Makes DRC-01 and ASMT-10 items visible to Overdue / Due-in-7 and to the calendar.
Scheduled, unattended runs	A <code>pg_cron</code> job enqueues <code>PULL_NOTICES_BUNDLE</code> for every active client at 06:00 IST; the agent runner processes the queue with session reuse; jobs that need a CAPTCHA park as <code>needs_human</code> and the app shows them in one place (the Sync & alerts card) for a person to clear in a few clicks. The extension remains the fallback for ad-hoc pulls.	Alerts and the daily digest are only useful if the data is fresh every morning without someone remembering to click.
Change detection → events	An AFTER INSERT OR UPDATE trigger on <code>gst_notices</code> writes <code>notice_events</code> (captured, changed, removed). The alert queue function consumes them.	One event source for email, in-app bell and the activity feed.
Active flag respected	Sync All skips clients marked inactive on the Company List and reports them as "skipped" in the run summary.	Fewer CAPTCHAs, fewer failed logins on cancelled GSTINs.
Version handshake	The extension announces its version in the ready ping; the app warns when it is older than the minimum it expects.	Prevents a stale extension silently writing an old row shape.

10 Implementation roadmap

10.1 Sequencing logic

The order below follows the two judges' recommendation (§10.3) and differs from the author's first draft in one respect: the GST team's inbox comes before the redesigned screen. The reasoning: the first three weeks make the numbers right, stop the sync from destroying staff work, close the security holes that no email link should be sent across, and write the module's positions down. Alerting ships next because it is the cheaper half of the ask and it works on the existing lists once the data is trustworthy. The redesigned dashboard, work queue and drawer follow; the matter layer, unattended sync and retention, and MIS complete the module. Each phase is independently shippable and leaves the app better than it found it. Every phase carries a week of slack in its estimate.

Phase	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0 · Stabilise, secure, write it down																				
1 · Email the GST team																				
2 · Dashboard, work queue, drawer																				
3 · Litigation core (3a then 3b)																				
4 · Unattended sync, retention																				
5 · MIS, client reporting, polish																				

Columns are weeks. One developer working with Claude Code plus a half-day a week of partner review: 20 weeks, about 130 person-days including slack. Two developers finish in about 12 weeks because Phases 1 and 2 can overlap once Phase 0 is done, and 3b can overlap 4.

10.2 Phase plan

Phase 0 — Stabilise, secure, write it down

Weeks 1–3 · ~18 person-days

Goal: a Sync All never destroys staff work or serves as an attack path; every tile has one written definition and reconciles with the lists behind it; the module's positions exist on paper.

0.1 Upsert sync — migration adds `portal_key`, `first_seen_at`, `last_seen_at`, `portal_hash`, `deleted_at` and the unique index; `replaceNotices`, `replaceRefundApplications`, `replaceDrc03Filings` and `replaceCaseFolderItems` become upserts that send portal-owned columns only and soft-mark keys not returned; a run that returns zero rows never marks anything missing. Back-fill `portal_key`; dedupe case-task rows by ARN.

0.2 Failure path and job lifecycle — a failed pull writes only `client_sync_log` (action per step: login / notices / refunds / drc03, with a reason class) and never a notice row; delete the five "PULL FAILED" writes; login failures are detected and logged; a top-level catch advances the job instead of stalling it; the 20-minute job expiry becomes idle-based so a 92-client run can finish; the extension announces its version and the app enforces a minimum; every bridge message carries the acting user.

0.3 One canonical notice set — a shared `useNoticeSet()` hook (fetchAllRows with an explicit `ORDER BY id`, errors surfaced, sentinel and soft-deleted rows excluded) feeding the tiles, Notice Summary, GSTIN-wise count and the all-clients list. Definitions: Open = not closed; Overdue = open, no reply logged, effective due date (extended or original, parsed as an IST calendar date) before today; Due-in-7 = open, effective due date within 0–7 days; New = `first_seen_at` within 24 h. A Vitest fixture pins these definitions and the refund / DRC-03 status vocabularies observed on the live portal.

0.4 Closing sweep and due dates — rule-based auto-close for LUT acknowledgements, the 131 DRC-03 acknowledgements and the 22 refunds already closed by an RFD-06 / PMT-03 order that the live check found counted as "open", and case rows whose folder holds a closure item; a bulk "Close with reason" action for the rest. Case/task rows get their due date and hearing date from the folder JSON the extension already stores (159 due dates and 35 hearing dates sit unread today); RFD-08 reply dates are kept instead of dropped.

0.5 Security stop-gaps (hand-verified, §4.2) — add `https://gst.vjdesai.com/*` to the extension manifest so Sync All works in production; make the app bridge accept messages only from the app's own origin; apply the role from `user_roles` on session restore instead of trusting the localStorage copy; guard `/clients`, `/add-client` and `/edit-client` for staff roles; make the `return-pdfs` bucket private, drop the "anyone can upload / delete" storage policies and serve PDFs through short-lived signed URLs; mask portal passwords everywhere and stop copying them into `chrome.storage`. Credential encryption and Supabase Auth adoption remain a separate project-wide track (§11.4).

0.6 Quick fixes — top-nav and sidebar active states from the route; aria-labels and keyboard access on icon buttons, calendar days and clickable cells; calendar "Both" contrast; real error states; no full-page reload after Import Company; inline status/priority edits that survive re-renders; hooks moved above early returns; the always-empty "Additional Notices" report removed or wired.

0.7 Write it down — `docs/NOTICES_LITIGATION_POSITIONS.md` recording the category taxonomy, the tile definitions, the refund / DRC-03 dedupe and closure rules, the sync constants and the CAPTCHA posture; a notices section in `CLAUDE.md`; a one-page staff SOP for Sync All and a runbook for a failed client login; `extension/README.md` brought up to date.

- ✓ Sync the same client twice: `staff_status`, `priority`, `assign_to`, `reply/order` fields unchanged; refund documents intact.
- ✓ No row in `gst_notices` has a description starting "PULL FAILED"; no client is left without history after a failed pull.
- ✓ Total / Open / Overdue on the dashboard equal the row counts of their drill-down lists and of the GSTIN-wise report; the Vitest fixture passes.
- ✓ Sync All starts on `gst.vjdesai.com`; a client-role login cannot open `/clients` or read a portal password; a notice PDF URL fetched without a session returns 400/403.
- ✓ `npm run build` green; the existing routes unchanged.

Phase 1 — Email every important matter to the GST team

Weeks 4–6 · ~16 person-days

Goal: the matrix in §6 is live on the corrected data; nobody on the team needs to open the app to learn that a notice arrived, is due, or was missed.

1.1 Event source — `notice_events` table and trigger on `gst_notices` (captured / changed / removed / assigned / status); staff actions write events from the app.

1.2 Rules, queue and outbox — `notice_alert_rules`, `notice_alert_log`, `staff_notification_prefs`; new edge function `queue-notice-alerts` on a 15-minute cron plus the 09:30 IST digest and Monday MIS runs; `email_outbox` extended (new kinds, nullable `client_id`, to/cc arrays, attachments, dedupe_key, a `preview` status for dry runs); `send-gst-email` cadence raised to every 15 minutes with attachment support and a "notice" variant of the corporate shell.

1.3 A named owner — `assign_to_user_id` replaces free-text `assign_to` (back-filled by name match); a staff picker in the list; "Mine" filter; the ladder escalates to a person, then the team list, then the partner.

1.4 Early statutory clocks — a minimal `matter_deadlines` table seeded from existing rows without waiting for matters: appeal (s.107) and tribunal (s.112) clocks from order dates and ORDRS folder items, rectification (s.161), RFD-08 reply dates, DRC-22 one-year attachments. E5 limitation alerts fire from this phase; Phase 3 re-parents these rows to matters.

1.5 Templates and a fatigue-safe rollout — E1–E11 seeded in `email_templates` under a "Notice alerts" tab of the Reminders page; week one runs digest-only with immediate alerts in preview status so the team can read what would have been sent; then per-client batching, an overdue repeat cap, quiet hours and the critical-only exception switch on.

1.6 Client-facing — E12 / E13 templates and a "Request documents from client" action, reusing the client reminder ladder and the per-client on/off switch.

1.7 Visibility — every send appears on the all-clients list (per-row "last alert") and in the outbox log; failures surface on the dashboard. Deep links land on `/notices-all?ref=...` until the drawer exists in Phase 2.

- ✓ A new DRC-01 captured by sync reaches `gst@vjdesai.com` within 15 minutes with the PDF attached.
- ✓ Re-running the cron sends nothing twice (`dedupe_key`); the preview week produces zero unexpected sends.
- ✓ The digest arrives at 09:30 IST every day the data changed; the partner receives the Monday MIS; an order with no appeal filed produces a T-30 limitation email.

Phase 2 — Dashboard redesign, work queue and matter drawer

Weeks 7–10 · ~26 person-days

Goal: the page in §5 — a partner opens it and sees what needs action today, who owns it and what money is at stake; every email deep-links into it.

2.1 Shell — new header (sync line, global search, actions), route-driven tabs, URL-persisted filter chips; page root on the app's 8-point rhythm; Company onboarding moved to the Clients page; one react-query hook over a Postgres view / RPC returning aggregates plus the top-N queue instead of four effects pulling the whole table.

2.2 Needs-attention strip — four tiles with the Phase 0 definitions plus ₹ exposure from `amount_of_demand`; each tile is a saved filter of the queue.

2.3 Work queue and permissions — sortable, filterable table with Team / Mine / Unassigned / Hearings tabs; bulk assign, priority, stage, email-team, export with confirmation and an audit event; permission keys `manage_notices`, `close_matters`, `partner_review`, `view_exposure`; sensitive writes moved behind SECURITY DEFINER RPCs that validate the app session.

2.4 Charts — ageing buckets, demand-by-stage donut, category bars, top exposure by client (recharts, semantic tokens, click-to-filter).

2.5 Next 14 days and Sync & alerts cards — deadline strip from `matter_deadlines`, month toggle, sync health with last-sync time and failed logins, activity feed from `notice_events`.

2.6 Matter drawer v1 — the drawer from §5.4 on a single notice: badges, facts, activity, documents (portal PDF, case-folder attachments, uploads, all via signed URLs), notes, log reply / order / submission with close-with-reason and reopen; replaces the current edit dialog; `/notices/:id` route for deep links.

2.7 Refunds and DRC-03 as typed, rupee-bearing data — cause of payment, period and SCN linkage kept by the parser; DRC-03 amounts back-filled from the case folders already stored (₹ 6.46 crore across 130 rows shows as zero today); `sanctioned_amount` populated from RFD-05 / 06; all captured documents reachable; one definition of "closed" for a refund case.

2.8 Responsive and accessible — one-column layout below 1024 px, card list on phones, focus-visible everywhere, contrast checked.

- ✓ Every number on the page opens a list whose count matches it.
- ✓ A notice can be assigned, prioritised, staged and closed with a reason without leaving the dashboard.
- ✓ Lighthouse accessibility ≥ 95 ; no text below 12 px; first paint under 1.5 s with 5,000 notices.

Phase 3 — Litigation core (3a: matters, deadlines, hearings · 3b: demand, documents, review, client view)

Weeks 11–15 · ~36 person-days

Goal: disputes are managed as matters through their statutory lifecycle, with deadlines, hearings, demand and payments in one place, and the client kept informed.

3a.1 Schema and back-fill — tables in §8.1; `litigation_rules` seeded from §7.1; one-off grouping of existing notices into suggested matters (by case ARN first) with a review page for staff to confirm; Phase 1 deadlines re-parented.

3a.2 Matters page — list and board views by stage; filters by lifecycle, client, owner, officer, FY; matter creation from any notice; stage stepper with computed next step and deadline; stage history; typed manual entry for lifecycles the portal does not expose (appeals, GSTAT, recovery, e-way bill, summons).

3a.3 Deadline engine and hearings — computed vs overriding deadlines with their statutory basis; hearings table, Hearings page with a calendar, ICS export, adjournment logging, documents-to-carry checklist; E4 reminders switched to hearing rows; appeal and rectification clocks created automatically on DRC-07 / APL-04 capture.

3b.1 Demand, payments and outcomes — demand table per Act and head; DRC-03 linkage as payments; pre-deposit calculator from `litigation_rules`; recovery / stay status; closure rules with outcome (dropped, confirmed, partly, won, lost) so win-rate and time-to-close exist.

3b.2 Documents, reply workspace and review — unified document list with kinds in the private bucket; reply drafting area with firm templates and links to precedent matters; partner review step recorded as an event; authority and counsel masters.

3b.3 Client view — read-only "Notices & litigation" page in the client portal showing stage, next date and documents requested; E12 / E13 wired to it.

- ✓ A DRC-07 captured today automatically shows an appeal last-date and the pre-deposit amount.
- ✓ Every open matter has an owner, a stage and a next date, or appears in an exceptions list.
- ✓ Exposure under dispute on the dashboard equals the sum over open matters; DRC 03 "open" counts only genuinely pending payments.

Phase 4 — Keep it running: unattended sync, retention and recovery

Weeks 16–18 · ~18 person-days

Goal: data is fresh every morning without anyone clicking, failures are visible and specific, and litigation records cannot be lost by a click or a cascade.

4.1 Agent handler — `PULL_NOTICES_BUNDLE` in `agent/src/handlers.ts` porting the notices, refunds, DRC-03 and case-folder capture from `content.js` to Playwright, using the same upsert path; `portal_agent` migration verified on the live project; `portal_jobs` writable only through an RPC.

4.2 Scheduling — `pg_cron` enqueues the bundle for active clients at 06:00 IST; retries with backoff on transient failures; `needs_human` CAPTCHA queue surfaced on the Sync & alerts card; run summaries with row counts.

4.3 Guardrails — portal-change detection (row shape or zero-row anomalies raise E9), active-flag respected, per-client credential health, session cookies with expiry and rotation.

4.4 Retention and recovery — soft delete on every litigation table; client delete becomes archive-with-audit and offers a non-destructive alternative; orphaned PDFs cleaned up; a documented and tested backup / point-in-time-recovery procedure; a legal-hold flag for matters under dispute (s.36 record-keeping).

- ✓ For 10 consecutive days, 100% of active clients have a sync younger than 24 h or a named failure reason.
- ✓ Zero human CAPTCHA entries on days when sessions are still valid.
- ✓ A deleted client can be restored with every notice, document and log intact.

Phase 5 — MIS, client reporting and polish

Weeks 19–20 · ~14 person-days

Goal: the partner and the client each get the report they need without a spreadsheet.

5.1 MIS — exposure by client / stage / officer / section; ageing; opened vs closed; outcomes and time-to-close; per-staff load; all exportable and attached to the Monday email.

5.2 Client litigation status report — per-client PDF on the firm's letterhead (existing report kit) listing open matters, stage, next date, demand and payments; one click from the client profile and from the drawer.

5.3 Polish and options — GSTIN-wise report on the canonical set; in-app notification bell over the same events; optional WhatsApp channel, PDF-to-fields extraction for manual notices, and AI-assisted first-draft replies evaluated as separate decisions.

- ✓ MIS figures match the dashboard; client report generated in one click and matches the drawer.

10.3 How the sequencing was chosen

Three drafts were written from different starting points and scored by two judges (1–10 per criterion).

Judge	Draft	Complete	Sequencing	Feasible here	Value to firm	Total
ca-partner	fix-first	9	8	7	9	33
ca-partner	user-first	9	8	7	8	32
ca-partner	platform-first	8	6	6	7	27
lead-engineer	fix-first	10	8	8	9	35
lead-engineer	user-first	9	8	8	9	34
lead-engineer	platform-first	9	6	7	7	29

What the judges agreed on. The partner judge preferred the fix-first draft as the spine (stabilise, email the team, rebuild the screen, litigation core, durability); the engineer judge preferred the user-first draft (stabilise plus security, the 08:30 inbox, the screen with owners, litigation core, refunds and MIS, keep-it-running). Both orders put the inbox before the redesign and both judges made the same four corrections: give Phase 0 three weeks rather than two, seed a minimal deadlines table from existing rows so limitation reminders fire in the email phase, roll alerts out digest-first with a preview week and caps, and pull the refund / DRC-03 rupee back-fill forward so the redesigned dashboard never shows ₹ 0 for a paid DRC-03. The engineer judge added the job-lifecycle items (idle-based expiry, login-failure logging, version handshake, acting user on every bridge message) to Phase 0 and asked for sensitive writes to move behind SECURITY DEFINER RPCs; the partner judge asked for Phase 3 to be split into 3a and 3b and for a week of slack per phase. Section 10.2 is that merge. The judges' full notes are in the review record.

11 Acceptance criteria, KPIs, risks and decisions

11.1 Day-one measurements (before any change)

Run once against the live project and record in the Phase 0 ticket, so the "after" can be proven: count of `gst_notices` rows; rows with description starting "PULL FAILED"; rows with a non-null `staff_status`, `priority`, `assign_to`, `reply_date`; duplicate (`client_id`, `reference_number`) pairs; case rows with null `due_date`; clients with a failed latest sync; clients whose latest sync is older than 7 days; `email_outbox` rows by kind and status in the last 30 days.

11.2 Acceptance criteria by phase

Phase	Done means
0	Two consecutive syncs of one client leave every staff column unchanged; no sentinel rows; tile counts equal their drill-down counts and the GSTIN-wise report; Sync All starts on <code>gst.vjdesai.com</code> ; a client-role login cannot read portal passwords or fetch a PDF without a session; positions document merged; build green; no route removed.
1	E1-E11 live on corrected data; a new DRC-01 reaches the team inbox within 15 minutes with the PDF; re-running the cron sends nothing twice; digest daily at 09:30 IST; Monday MIS to partners; templates editable by staff; an order with no appeal filed produces a T-30 limitation email.
2	Partner can, from the dashboard alone, find today's overdue items with demand, assign them, set priority and close with a reason; DRC-03 and refund rows show real amounts; Lighthouse accessibility ≥ 95 ; no text under 12 px; page usable at 1024 px and on a phone.
3	All open notices attached to a matter or listed as exceptions; DRC-07 capture creates an appeal clock and pre-deposit figure automatically; hearings on the calendar with reminders; client view live for one pilot client.
4	Ten consecutive days with every active client synced within 24 h or a named failure reason; CAPTCHA queue cleared in-app; run summaries on the dashboard; a deleted client restorable with all records; backup / recovery procedure tested once.
5	Exposure, ageing, outcome and load reports match the dashboard; client status PDF in one click; first paint under 1.5 s with 5,000 notices.

11.3 KPIs the partners should watch monthly

- Missed statutory deadlines (target: zero; every miss gets a written cause).
- Replies filed at least 3 days before the due date (target: 90%).
- Notices assigned within 24 hours of capture (target: 100%).
- Median time from capture to first action (target: under 2 working days).
- Sync freshness: active clients synced in the last 24 h (target: 100%).
- Staff-data-loss incidents (target: zero, verified by the Phase 0 test).
- Exposure under dispute and pre-deposits outstanding, by client and stage.
- Outcomes: matters dropped / reduced / confirmed, and time-to-close by lifecycle.

11.4 Risks and mitigations

Risk	Mitigation	Owner / phase
Portal API or DOM changes break capture silently	Zero-row and shape anomalies raise E9; extension/agent version handshake; a "shadow" run mode before switching writers.	Phase 4; reviewed each portal release
GSTN terms on automated access	No CAPTCHA solving; session reuse with a human in the loop; written client authorisation; agent machine office-local; credentials encrypted at rest.	Partner decision before Phase 4
Client credentials, notice PDFs and every litigation row readable and writable with the public anon key (open RLS by project rule; public storage bucket; role trusted from localStorage)	Phase 0 stop-gaps (private bucket and signed URLs, storage policies, route guards, role from the database, bridge origin check, password masking); then a project-wide track: credentials behind an edge function with the service role, encryption at rest, Supabase Auth for staff so RLS can be real, sensitive writes through SECURITY DEFINER RPCs.	Phase 0 now; security track decision
Migrating 1,060 rows into matters mis-groups disputes	Suggested-matters review page; grouping by case ARN first; nothing auto-merged without a staff confirmation.	Phase 3a
Records lost by a client delete or a cascade; no backup procedure documented	Soft delete on litigation tables, archive-with-audit instead of delete, PDF cleanup, tested point-in-time recovery, legal-hold flag.	Phase 4
Institutional knowledge lives in code comments; git history was squashed	Positions document, CLAUDE.md section, staff SOP and runbook in Phase 0; every later position change edits the document first.	Phase 0
Alert fatigue	Digest-only preview week, team list plus roles, dedupe keys, per-client batching, overdue repeat cap, quiet hours, critical-only immediate sends outside hours; rules editable by staff.	Phase 1
Statutory defaults drift with Finance Acts	<code>litigation_rules</code> table with effective dates; computed vs overriding deadlines; every alert names its basis.	Phase 3; annual review
No automated tests in the project	Phase 0 adds a small Vitest suite for the tile definitions, date handling and the upsert contract; every phase adds tests for its engine code.	Phase 0 onward

11.5 Decisions requested from the partners

1. Approve the legal posture for unattended sync (§9.1 box) or restrict Phase 4 to extension-only runs.
2. Name the distribution list for team alerts and the partner recipients for escalations and the Monday MIS.
3. Confirm auto-close rules: LUT acknowledgements, DRC-03 acknowledgements, case rows with a closure item.
4. Confirm the statutory defaults in §7.1 as the firm's positions, in the same way the Builder and Advance positions are documented today.

5. Approve the Phase 0 security stop-gaps now, and decide on the project-wide security track in §11.4 (credential encryption, Supabase Auth for staff, RPC-guarded writes).
6. Confirm that alerting (Phase 1) precedes the redesign (Phase 2), as both judges recommended, or reverse it if the visual change is the more urgent need this quarter.
7. Staffing: one developer for 20 weeks or two for about 12.

A Appendix A · Gaps and improvements catalogue (123 items)

Everything the review found missing or improvable, ordered by priority. These are design judgements rather than defects; each names the proposal and the effort. The roadmap in §10 schedules the critical and high items; medium and low items ride along with the phase that touches their area.

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G1	■ Critical	data-model	No Matter entity: every SCN, reminder, order and acknowledgement is an unrelated row A DRC-01A intimation, its DRC-01 SCN, the reminder, the DRC-07 order and the APL-01 appeal are one dispute for the firm but 4-5 unrelated rows here; totals (1060) and Open (1060) count documents, not matters, and no ...	CREATE TABLE gst_matters (id uuid PK DEFAULT gen_random_uuid(), client_id uuid NOT NULL REFERENCES clients(id) ON DELETE CASCADE, matter_no text UNIQUE, title text NOT NULL, matter_type text NOT NULL CHECK (matter_type IN ...	XL
G2	■ Critical	data-model	No per-employee data scoping exists anywhere — a 'my matters' queue has nothing to reuse; assign_to is free text A CA litigation practice runs on allocation: each SCN/ASMT-10/DRC-01 is owned by a named senior with a manager reviewing. Without an assignee FK there is no 'what needs my attention today', no workload view per person, ...	New migration: `ALTER TABLE gst_notices ADD COLUMN assigned_to uuid REFERENCES profiles(user_id), ADD COLUMN reviewer_id uuid, ADD COLUMN assigned_at timestamptz;` keep assign_to as a display fallback and backfill by matching ...	L
G3	■ Critical	email-alerts	No 'new notice landed' event or email — a notice can sit unseen until someone opens the dashboard The firm's stated goal is 'emailing everything to GST team for every important matter'. A show-cause notice with a 7-day reply window that lands during a sync no one reviews is exactly the litigation miss this ERP ...	New table `gst_notice_events` (id uuid pk, notice_id uuid references gst_notices on delete cascade, client_id uuid, event_kind text check (event_kind in ...	L
G4	■ Critical	email-alerts	No email alert when a new notice is pulled or a due date approaches (every benchmark product has this) 473 of 1060 notices are already overdue on the screenshot. Nobody in the GST team is told when a notice lands or when a reply window is closing unless they open this page; a missed DRC-01 reply window converts to an ...	Extend the existing pipeline rather than build a new one: (a) widen `email_outbox.kind` CHECK to add 'notice_new','notice_due','notice_overdue','notice_digest'; (b) add `notice_id uuid` to email_outbox; (c) new edge function ...	L

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G5	■ Critical	email-alerts	Prerequisites before notice e-mails carry PDF links — ranked change list The principal's ask is to e-mail the GST team and clients on every important matter. E-mails are forwarded, quoted and archived; a permanent public URL in one means the demand order is readable by anyone who ever sees ...	Ranked, in order: 1) make return-pdfs private + strip /object/public/ prefixes from stored URLs + signed-URL edge function (finding 3); 2) drop 'Anyone can delete/update return PDFs' (finding 4); 3) origin/source check in appbridge.js + ...	L
G6	■ Critical	litigation-gap	No 'needs attention today' queue - the dashboard shows counts only, never a single notice A litigation ERP's home screen must answer 'what do I reply to today, for whom, by when' without a second click. Today the GST team has to open a 1060-row list and filter by hand, so overdue replies slip.	Add an 'Action queue' card (col-span=8) below the tiles: tabs Overdue / Due in 7 days / Unassigned / High priority / New since last sync; columns Client, Category chip, Description, Due (days badge), Owner select, Amount, Priority, actions ...	L
G7	■ Critical	litigation-gap	No positions document for the notices/refunds/DRC-03 module — every elected position lives only in code comments Every other audit dimension proposes rewriting the tiles, taxonomy, Open/Closed logic and sync flow. Without a register of what the firm already decided (and why), the rewrite discards positions that were confirmed live ...	Create docs/NOTICES_LITIGATION_FLO W.md, structured like docs/2B_RECONCILIATION_FLO W.md (a 'positions were implemented by engineering judgement, not a sign-off' preamble, a flow diagram, numbered § positions, a 'Known limitations' and a ...	M
G8	■ High	data-model	No audit trail of who changed what on a notice — audit_log covers only 'clear_data' When a notice is found Closed with no reply on file, nobody can tell whether a staff member closed it or a sync wiped the status. For a CA firm, the litigation file's change history is itself evidence of due diligence.	CREATE TABLE gst_matter_activity (id bigserial PK, matter_id uuid REFERENCES gst_matters ON DELETE CASCADE, notice_id uuid, table_name text NOT NULL, row_id uuid NOT NULL, action text NOT NULL CHECK (action IN ...	M
G9	■ High	data-model	No new-vs-updated detection or change log for notices The firm cannot answer 'what came in since yesterday', 'which due dates were extended', 'which SCN got an order' without opening the portal. Every email alert, escalation and audit trail for litigation depends on ...	Create gst_notice_events(id, client_id, notice_id, reference_number, event text check in ('new','due_date_changed','status_changed','order_received','pdf_captured','missing_from_portal'), old_value jsonb, new_value jsonb, sync_run_id uuid, ...	M

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G10	High	data-model	No staff-side document store on a notice (reply drafts, evidence, working papers) The reply, its annexures, the DRC-06 acknowledgement and the client's supporting invoices live in email and Drive today; the case folder page shows only what the portal returned.	INSERT INTO storage.buckets (id,name,public) VALUES ('litigation-docs','litigation-docs',false); CREATE TABLE gst_documents (id uuid PK DEFAULT gen_random_uuid(), client_id uuid NOT NULL REFERENCES clients ON DELETE CASCADE, matter_id uuid ...	M
G11	High	data-model	staff_status is free text with only Open/Closed in the UI; no litigation lifecycle states A CA firm's litigation pipeline is New → Under Review → Reply Drafted → Reply Filed → Hearing Scheduled → Order Received → Appeal Filed / Closed / Dropped. With two states the firm cannot see where each matter sits, who ...	Migration: CREATE TYPE notice_status AS ENUM ('new','under_review','awaiting_client','reply_drafted','reply_filed','hearing_scheduled','order_received','appeal_filed','closed','dropped'); ALTER gst_notices ADD COLUMN status_v2 ...	M
G12	High	data-model	Closing/reopening a notice needs no reason, no order reference, and leaves no trace A partner cannot tell who closed a notice, when, or why (order accepted? demand paid via DRC-03? appeal filed?). Closed can be flipped back to Open by anyone with edit_notice_status with no record. This fails basic ...	Add table gst_notice_events (id, notice_id FK, client_id, event_type text, from_value, to_value, note, actor_user_id, actor_name, created_at) written from a small src/lib/noticeEvents.ts helper on every status/priority/assign/field change ...	M
G13	High	data-model	No history/snapshot or restore for notice, refund, DRC-03 and case-folder rows A staff member overwriting a reply date, a bulk status change to the wrong 20 rows (NoticeWorkflowListView.tsx:435 bulk update), or a sync that drops a row leaves no way to see the prior value or bring it back. For a ...	One migration: `gst_litigation_history (id, table_name text, row_id uuid, client_id uuid, op text check (op in ('UPDATE','DELETE','SYNC_DELETE')), old_row jsonb not null, actor text, reason text, changed_at timestampz default now())` with ...	M
G14	High	data-model	No backup, PITR or recovery procedure is documented or tested anywhere in the repo Every other finding in this dimension assumes a last-resort restore exists. Whether Supabase project gcquafqxbykxbexcdpy has daily backups or PITR enabled is not visible from the code and nobody in the firm has a ...	Add docs/BACKUP_AND_RECOVERY.md: confirm the Supabase plan and enable PITR (minimum 7 days) on project gcquafqxbykxbexcdpy; schedule a nightly pg_dump of the litigation tables plus a bucket mirror to an off-platform store (e.g. Google ...	M

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G15	High	data-model	DRC-03 cause-of-payment code, period and SCN linkage are discarded by the parser Whether a DRC-03 was voluntary u/s 73(5)/74(5), against an SCN, an audit para or a pre-deposit decides whether the linked notice can be treated as discharged and whether penalty exposure remains; the firm cannot see ...	Add cause_code, cause_text, scn_reference, scn_date, payment_ref (prn), period_from/to from ovtp to gst_drc03_filings; resolve scn_reference to gst_notices.reference_number to set a notice_id FK and show 'Paid via DRC-03 ARN ...' on the ...	M
G16	High	email-alerts	Email pipeline cannot carry a notice — kind CHECK and templates have no notice/litigation kind and no notice_id The principal's core ask — 'emailing everything to the GST team for every important matter' — has no schema path: an INSERT with kind='notice_alert' is rejected by the CHECK, and even a hand-crafted row could not be ...	ALTER TABLE email_outbox DROP CONSTRAINT email_outbox_kind_check, ADD CONSTRAINT email_outbox_kind_check CHECK (kind IN ...	M
G17	High	email-alerts	No due-date escalation ladder for notices (T-7 / T-3 / T-1 / T0 / overdue) — the 473 overdue rows never trigger anything The screenshot shows Over Due 473 of 1060 and 7 Days Due 0. Missing a reply deadline under s.73/74 converts a notice into an ex-parte order; the firm's exposure is the sum of amount_of_demand on those rows and nobody is ...	New edge function `queue-notice-reminders`, pg_cron `gst-queue-notice-reminders-daily` at `30 3 * * *` (09:00 IST). Query: `gst_notices where source='notices' and lower(coalesce(staff_status,'')) <> 'closed' and coalesce(extended_due_date, ...	M
G18	High	email-alerts	No daily digest to the GST team and no weekly portfolio summary to partners Partners and the GST team currently only learn the state of 1060 notices by logging in and looking at tiles; a 'what needs attention today' push is the single highest-leverage habit change for a litigation desk and the ...	Two edge functions. (a) `notice-daily-digest`, pg_cron `0 3 * * 1-6` (08:30 IST Mon-Sat): sections New since yesterday (first_seen_at > now()-1d), Due in 7 days, Overdue (sorted by amount_of_demand desc), Unassigned open, Hearings this ...	M
G19	High	litigation-gap	No owner / assignee dimension anywhere on the dashboard The principal cannot see who owns which notice, who is overloaded, or which notices have nobody assigned - the basis for delegation and for emailing the responsible person.	Migrate assign_to to assigned_to uuid REFERENCES users(id) (keep the text column during transition), add 'Unassigned' KPI tile, a 'Workload by owner' bar chart (open/overdue per staff) and an owner filter in the filter bar; queue rows get ...	M

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G20	High	litigation-gap	No stage/status state machine — staff_status is free text with only 'closed' recognised The dashboard can only say Open or Closed; it cannot show 'reply drafted / awaiting client docs / PH fixed / order awaited / appeal filed', cannot report cycle time per stage, and cannot alert the GST team when a matter ...	On gst_matters: stage text NOT NULL DEFAULT 'new' CHECK (stage IN ...	M
G21	High	litigation-gap	No statutory limitation dates — only portal due_date/extended_due_date; 'Over Due' = due_date < today 473 'overdue' items are mostly closed/replied notices whose portal due date passed — noise that hides the handful of real limitation deadlines. Missing an appeal limitation is unrecoverable for the client and an ...	CREATE TABLE gst_matter_deadlines (id uuid PK, matter_id uuid NOT NULL REFERENCES gst_matters ON DELETE CASCADE, notice_id uuid REFERENCES gst_notices ON DELETE SET NULL, kind text NOT NULL CHECK (kind IN ...	M
G22	High	litigation-gap	No hearings/adjournments table Personal hearing dates are the single most time-critical event in a GST dispute; today they live in someone's diary, so the dashboard calendar (issue/due only) cannot show them and no email can remind the GST team or ...	CREATE TABLE gst_hearings (id uuid PK DEFAULT gen_random_uuid(), matter_id uuid NOT NULL REFERENCES gst_matters ON DELETE CASCADE, notice_id uuid REFERENCES gst_notices ON DELETE SET NULL, scheduled_on timestampz NOT NULL, venue text, ...	M
G23	High	litigation-gap	Demand is a single number — no tax/interest/penalty × CGST/SGST/IGST/cess breakdown, no admitted vs disputed vs paid The firm cannot report total disputed demand by client/FY/forum, the amount admitted and paid, or the contingent liability figure auditors ask for; pre-deposit for an appeal (10% of disputed tax) cannot be computed from ...	CREATE TABLE gst_demand_lines (id uuid PK, matter_id uuid NOT NULL REFERENCES gst_matters ON DELETE CASCADE, notice_id uuid REFERENCES gst_notices ON DELETE SET NULL, document_stage text CHECK (document_stage IN ...	M
G24	High	litigation-gap	DRC-03 payments are not linkable to the notice/matter they discharge (pre-deposit, payment against order) 'Has the 10% pre-deposit for this appeal been paid?' and 'how much of this demand has already been paid voluntarily?' cannot be answered; ITC-vs-cash split per matter is lost; DRC-03 rows are shown as their own 'notice' ...	ALTER TABLE gst_drc03_filings ADD COLUMN matter_id uuid REFERENCES gst_matters(id) ON DELETE SET NULL, ADD COLUMN notice_id uuid REFERENCES gst_notices(id) ON DELETE SET NULL, ADD COLUMN payment_purpose text CHECK (payment_purpose IN ...	S

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G25	High	litigation-gap	Demand vs paid ledger per matter is impossible: one numeric amount_of_demand, no head/FY/stage split, no DRC-07 amounts captured The partner cannot answer 'what is our clients' total exposure', 'how much was confirmed vs dropped in the order', or 'what is still recoverable' — the core numbers of a litigation practice.	Table matter_demands (matter_id, stage [proposed_drc01a, scn_drc01, order_drc07, apl04, gstat, hc], financial_year, head [IGST,CGST,SGST,Cess], tax, interest, penalty, other, status [proposed, confirmed, dropped, set_aside, remanded]); ...	L
G26	High	litigation-gap	No partner review / approval workflow; edit_notice_status is the only permission A junior can close a matter, change an amount of demand or record a reply as filed with no maker-checker; replies and appeal grounds go out without a documented partner sign-off — a professional-liability exposure for ...	Table matter_approvals (matter_id, subject [reply_draft, appeal_grounds, close_matter, settle_amnesty], requested_by, approver, requested_at, decided_at, decision, comments); new permissions review_litigation and approve_litigation_filing ...	M
G27	High	litigation-gap	Clients cannot see their notices and there is no one-click 'notify client' The client is the one who must produce documents and pay pre-deposits; today staff forward PDFs by hand and clients call to ask status.	Seed template `client_notice_intimation` (kind 'client_notice', audience 'client'): 'We have received {{notice_type}} {{reference_number}} dated {{issue_date}} for GSTIN {{gstin}}; reply due {{due_date}}; please share the following by ...	M
G28	High	litigation-gap	No retention policy or legal-hold concept for GST notices and orders (s.36 CGST) Section 36 CGST requires books and records to be retained until the expiry of 72 months from the due date of the annual return for the year, and a taxpayer who is a party to an appeal, revision or any proceeding must ...	Add `retention_until date` (generated: annual-return due date for financial_year + 72 months) and `legal_hold boolean default false` to gst_notices/gst_refund_applications/gst_drc03_filings; set legal_hold automatically when staff_status ...	M
G29	High	litigation-gap	Client role has no notice visibility at all — no 'my notices' page, no sidebar entry, no dashboard tile For a litigation ERP the client is a party to the matter: they need to see the notice, its due date, what the firm filed (submission ARN/date at gst_notices.submission_arn), the hearing date and the order, and to ...	Add a client route (e.g. /my-notices) rendering NoticeWorkflowListView in read-only mode (canEdit false) fed by `gst_notices` where client_id = user.id and source='notices' plus the client's refunds/DRC-03, hide ...	M

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G30	High	litigation-gap	CLAUDE.md has no notices/litigation section, unlike Builder and 2B Claude Code sessions (the only change path per CLAUDE.md) are told to read positions docs before touching Builder and 2B, but get no such instruction for a module whose rules were reverse-engineered live against the GST ...	Add this section to CLAUDE.md after the 2B section: ## Notices Dashboard → Refunds / DRC-03 → Additional Notice Folder (litigation module) A firm-wide litigation tracker modelled on Notice Alert (noticealert.microvistatech.com) : ...	S
G31	High	portal-sync	Lifecycle capture gaps: appeals, GSTAT, recovery, e-way bill, DRC-01B/01C, REG-17/21, SPL-01/02, summons are neither pulled nor enterable in a typed way An appeal filed on the portal (APL-01 → APL-02 ack → APL-04 order), a GSTAT appeal (separate gstat.gov.in portal), a DRC-13 garnishee, an MOV-09 penalty order, a REG-17 SCN with 7-day reply, or a summons received ...	(1) Extension roadmap: enumerate case/search caseTypeCd values beyond RFUND/ADJVP (appeal 'APL', audit 'ADT', recovery, registration) and pull them into matter_events; (2) typed manual-entry forms per matter_type (Appeal form with ...	XL
G32	High	portal-sync	No unattended/headless notice sync exists: agent has no notices job, nothing enqueues jobs, schedule disabled, CAPTCHA unsolved Today one human must solve one CAPTCHA per client per run (92 CAPTCHAs for a full sync), so syncs happen rarely and the dashboard is chronically stale. The building blocks for a nightly job exist but are not wired to ...	(1) Add PULL_NOTICES_BUNDLE to the agent that calls the same JSON endpoints the extension already proved (services/auth/api/get/notices, litserv/auth/api/case/task/get, case/folder, case/search) via page.request with the saved session, ...	XL
G33	High	portal-sync	Notice sync is manual, per-client CAPTCHA-gated and has no freshness indicator 'Last 24 Hours: 64' measures when someone last clicked Sync All, not when notices were issued; a client whose sync failed three weeks ago silently disappears from every KPI. GSTN's terms and the CAPTCHA make fully ...	(a) Add a 'Sync freshness' strip on the dashboard: clients never synced / synced > 7 days ago / failed last attempt, computed from 'client_sync_log' already fetched (lines 174-185), each linking to /notices-company-list pre-filtered; (b) ...	L
G34	High	portal-sync	No staff SOP for Sync All and no runbook for a failed client login / failed pull Sync All over 92 credentialed GSTINs means 92 CAPTCHAs typed by a person who must keep one tab open; the live client_sync_log already holds 'failed' rows and nobody on the GST team has written guidance on whether to ...	Add a 'Staff SOP' section to docs/NOTICES_LITIGATION_FLOW.md (and a short pointer in extension/README.md). SOP — Sync All (10 lines): 1. Use Chrome with the GST Keeper Portal Sync extension loaded (chrome://extensions → Developer mode → ...	S

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G35	High	reporting	No ageing, trend or category charts despite recharts and chart.tsx being in the app With 1060 notices the firm needs to see whether the backlog is growing, how old open matters are, and which categories drive the load - none of which a static count conveys.	New /litigation/mis page fed by a view v_litigation_mis over litigation_matters + matter_demands + matter_deadlines: ageing (0-30/31-90/91-180/180+ days since issue and since last action), exposure heat-map client x FY, by section and by ...	M
G36	High	reporting	No test harness; KPI and category definitions are unpinned and inline in the component Any rewrite of Open/Overdue/category rules will silently change partner-facing numbers with nothing to catch it, and even the auditors' own spot-check was bitten by the cap.	Minimal vitest setup: devDependencies `"vitest": "^2.1.9"` (Vite 5 compatible); script `"test": "vitest run";` in vite.config.ts add `//` <reference types="vitest" />` and `test: { environment: 'node', include: ['src/**/*.test.ts'] }` ...	M
G37	High	reporting	KPI-tile definitions (Last 24 Hours = pulled_at, Over Due ignores Closed, Open = staff_status only) are recorded nowhere and were never signed off The principal reads 'Open Notices 1060' and 'Over Due 473' as litigation exposure, but they are artefacts of definitions chosen to mimic Notice Alert's labels; the KPI/workflow dimensions will change them, and without ...	Write §3 'Tile definitions' in docs/NOTICES_LITIGATION_FLOW.md with one row per tile (source column, formula, filter param it drills into, tag). Tag Last 24 Hours (pulled_at), Over Due (ignores Closed), Open (staff_status only) and 'tiles ...	S
G38	High	reporting	No rupee KPIs: refund exposure and DRC-03 outflow never surface on the dashboard A litigation ERP for a CA firm must answer 'how much client money is stuck in refunds' and 'how much has been paid under protest' at a glance; today neither number exists anywhere except a drill-down TOTAL row that is ...	Add tiles 'Refunds pending (Rs)', 'Sanctioned not disbursed (Rs)', 'DRC-03 paid this FY (cash/credit)' and 'Interest & penalty paid', each linking to the filtered drill-down; compute in a Postgres view so the numbers are consistent across ...	M
G39	High	security	Accepted design: with RLS open to public, any logged-in client user (or anyone holding the anon key) can read and write the whole litigation dataset — quantified blast radius This is the documented CLAUDE.md position ("RLS open to public ... let login, staff-only routing and the permission keys be the gate"), so it is not a bug to patch with auth.uid(). But it means the security of every ...	Keep the open-RLS design if the firm accepts it, but (1) move gst_password out of the anon-readable row: store it encrypted (pgsodium / pgp_sym_encrypt with a key in Vault) and expose it only through a SECURITY_DEFINDER RPC ...	XL

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G40	High	security	Client GST portal passwords are stored and transmitted in plaintext to three consumers; README promises encryption at rest that does not exist The Sync All job copies 121 plaintext portal passwords into the extension's chrome.storage.local (readable from chrome://extensions > service worker devtools and from disk on the staff PC) every time it runs, on top of ...	Encrypt gst_password at rest (pgsodium column encryption) and decrypt only in a SECURITY DEFINER RPC; have background.js fetch one client's credentials just-in-time at its login step instead of embedding all of them in `gstk_active_job`, ...	L
G41	High	security	Any holder of the anon key can delete litigation rows and PDFs without logging in For returns and reconciliations the convention is a trade-off; for litigation evidence it means a former employee, a leaked extension zip, or a curl one-liner can erase a client's order history with no login, no ...	Without changing the app's auth model: split DELETE out of the FOR ALL policies on the four litigation tables and grant DELETE to nobody (all app deletes go through SECURITY DEFINER RPCs that require a valid `portal_sessions`-style staff ...	M
G42	High	security	Permission matrix for the notices/litigation surface as built today (one key, no scoping, gst_manager == superadmin) The firm cannot express the roles a litigation practice actually has (partner who signs off, manager who allocates, senior who drafts replies, article who only reads and collects documents, client who should see their ...	Target matrix (rows as above): partner(superadmin)=all; GST manager=all except Manage Employees/User Control unless granted; senior(employee)=view + sync_notices + edit_notice_status + close_notice + assign_notice + export_notices, ...	M
G43	High	security	Every UI gate on the notices surface can be bypassed with the anon key: RLS is FOR ALL TO public on gst_notices, case folders, sync log, clients and even user_permissions This is by design for the app's custom localStorage auth, and the principal should know its consequence for litigation data: an employee with no keys can, from the browser devtools, close/reassign/delete any notice, ...	Keep RLS public (per project rule) but move sensitive writes behind SECURITY DEFINER RPCs that take the staff user_id plus a server-issued session token (extend authenticate_staff to return one, store it alongside vjdesai_user) and check ...	XL
G44	High	ux	No detail page/drawer for a plain notice — most rows have no click target beyond a 23-column grid row To act on one notice staff must read a 280px-truncated description in a horizontally scrolling table, open the PDF in a new tab, come back, open the pencil dialog, save, then change the status Select separately. There ...	Add a NoticeDrawer (Sheet, right side, 560px) opened by clicking the row or a 'View' icon: header (client, ref, type, issue/due/extended due, days-to-due, demand, priority, assignee, status), tabs Details Case Folder (reuse ...	L

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G45	Medium	accessibility	Sub-caption type (10-11px) and low-contrast tints fail the design rules and WCAG AA The page is read all day by staff on 1366px laptops; 11px muted text at borderline contrast is fatiguing, and the colour-coded days are the only cue for due dates.	Adopt the doc's scale: body text-sm, captions text-xs, section titles text-lg; use full-strength tokens for the marker (a dot in bg-destructive, bg-info, bg-warning) with text-foreground for the digit.	S
G46	Medium	data-model	Reminders/tasks cannot reference a notice — staff_task_reminders uses client as free text A 'draft reply by Friday' task cannot appear on the notice, cannot roll up to the matter, and cannot be reassigned when the employee is on leave.	ALTER TABLE staff_task_reminders ADD COLUMN client_id uuid REFERENCES clients(id) ON DELETE SET NULL, ADD COLUMN notice_id uuid REFERENCES gst_notices(id) ON DELETE SET NULL, ADD COLUMN matter_id uuid REFERENCES gst_matters(id) ON DELETE ...	S
G47	Medium	data-model	No notes/comments thread — remarks is a single overwriteable text column Case notes accumulate over months (officer's oral observations, client's explanation, counsel's view); one text box means the previous note is lost and nobody knows who wrote what.	CREATE TABLE gst_matter_notes (id uuid PK DEFAULT gen_random_uuid(), matter_id uuid NOT NULL REFERENCES gst_matters ON DELETE CASCADE, notice_id uuid REFERENCES gst_notices ON DELETE SET NULL, body text NOT NULL, is_pinned boolean NOT NULL ...	S
G48	Medium	data-model	gst_case_folder_items has no FK to the notice, no unique key, typed columns are absent — delete-then-insert again The Additional Notice Folder page must parse raw_json client-side for every date, so the folder's reply-due and order dates can never feed the dashboard, deadlines or emails; a case whose notice row is deleted leaves ...	ALTER TABLE gst_case_folder_items ADD COLUMN notice_id uuid REFERENCES gst_notices(id) ON DELETE CASCADE, ADD COLUMN matter_id uuid REFERENCES gst_matters(id) ON DELETE SET NULL, ADD COLUMN item_type text, ADD COLUMN issue_date date, ADD ...	M
G49	Medium	data-model	No idempotency key for alerts — only filing confirmations are protected from duplicates The 15-minute event poller, the daily ladder and a manual 'Send queued now' will overlap; without a DB-enforced key the GST team gets the same T-3 email twice and stops trusting the channel.	`ALTER TABLE email_outbox ADD COLUMN dedupe_key text; CREATE UNIQUE INDEX uq_email_outbox_dedupe ON email_outbox(dedupe_key) WHERE dedupe_key IS NOT NULL;` Every notice function inserts with a deterministic key (`notice.new:{notice_id}`, ...	S

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G50	Medium	data-model	Authority is a free-text issued_by: no CGST vs SGST, no officer/range/division master, jurisdiction captured but never joined The firm cannot filter or report by Centre vs State, by Commissionerate/Division/Range, or by officer — which is how a practice decides who argues, which precedent applies and where the appeal lies (Appellate Authority ...	Master table tax_authorities (id, administration [CGST, SGST, DGGI, Appellate, GSTAT, HC], designation, office_name, commissionerate, division, range, state, email, address) + authority_id on litigation_matters; auto-suggest from issued_by ...	M
G51	Medium	data-model	Section of law and financial year are free text; multi-FY notices cannot be represented; portal sec/fy fields are discarded Most s.73/74 SCNs cover several FYs (e.g. 2017-18 to 2020-21) with different limitation dates per FY; without section + FY[] the firm cannot group by issue (17(5), 16(4), RCM, e-way bill), cannot run s.128A eligibility ...	Add section_of_law text[] and financial_years text[] (validated 'YYYY-YY' picker) to litigation_matters and gst_notices; a sync-time backfill reads sdtls.sec / sdtls.fy from gst_case_folder_items.raw_json; child table matter_periods ...	S
G52	Medium	data-model	Refund lifecycle stages (RFD-02/03 deficiency, RFD-08 SCN, RFD-09 reply, RFD-06 order, RFD-05/PMT-03) are not modelled and RFD-08 due dates never reach the KPIs An RFD-08 has a 15-day reply window and an unanswered one becomes an RFD-06 rejection; the dashboard's 7 Days Due tile cannot see it, and deficiency memos (which restart the 2-year limitation clock) are invisible.	Add refund stages to the matter stage machine (filed, acknowledged_rfd02, deficiency_rfd03, scn_rfd08, reply_rfd09, order_rfd06, payment_rfd05, recredit_pmt03, withdrawn) fed from gst_case_folder_items (APLCN/NOTAC/REPLY/ORDRS) at sync; ...	M
G53	Medium	data-model	Documents of one dispute are not chained (no parent/matter link), so dashboard totals count the same matter 3-4 times and no timeline exists Total Notices = 1060 mixes intimations, reminders and orders of the same disputes; the team cannot open one record and read the chronology from intimation to order to appeal.	Add matter_id to gst_notices; at sync, group by case_id/arn where present and otherwise by (client_id, section, FY, issuing officer) heuristics with a manual 'Link to matter' action in the list; dashboard toggle 'Count: Matters ...	M
G54	Medium	data-model	No structured tagging by section, issue type, state or FY — dashboard filter is only Registration / Other A firm handling 92 GSTINs across states needs 'all s.73 ITC-mismatch notices for FY 2019-20 in Gujarat' as a two-click filter for bulk strategy and for the audit-trail of a common ground of appeal.	Add 'section_invoked text', 'issue_tags text[]', 'tax_period_from/to date' to gst_notices; derive 'state' from GSTIN prefix (clients.gstin[0:2]) at query time. Replace the single select with a filter bar (FY, State, Section, Issue tag, ...	M

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G55	Medium	data-model	REFUTED duplicates claim; the real duplication is tasks-per-case, which inflates every category count A partner sees '17 appeals' when the firm has 4 appeal cases; per-case ageing, assignment and reply tracking cannot work while the unit is the task.	Model a `gst_notice_cases` view/table keyed on (client_id, case_id) with latest task, latest due date, ORDRS/CLOSR presence; count cases in Notice Summary and list tasks beneath each case in the drill-down.	L
G56	Medium	data-model	No actor is recorded on any sync write or sync-driven deletion When a client's notices vanish after a sync, the firm cannot tell which employee ran it from which machine. Combined with the open Sync All button this makes the destructive path unattributable.	Add `actor_user_id` uuid, `actor_name` text, `machine_text` to `client_sync_log`; have the SPA pass the logged-in `vjdesai_user` id and name in the `__gstkPullSectionAllClients` `postMessage` (NoticesDashboardPage.tsx:233, CompanyListPage.tsx:232) and ...	S
G57	Medium	data-model	No way to delete a wrongly added notice except deleting the whole company (which cascades all its notices) A manual Add Notice typo (wrong client, duplicate reference) permanently inflates Total/Open counts and the calendar; the only 'fix' available to staff is to mark it Closed, which still counts in Total and Closed and ...	Add `deleted_at` timestamp, `deleted_by` uuid to `gst_notices` (soft delete), filter `is('deleted_at', null)` in every reader (fetchAllRows calls in NoticesDashboardPage, AllClientsNoticesPage, NoticeSummaryReportPage, ...	S
G58	Medium	data-model	Refund and DRC-03 'closed' status vocabularies exist only as regexes — the observed portal status strings are recorded nowhere Whether a refund counts as closed when 'Sanctioned' but not yet 'Disbursed', or a DRC-03 when 'Pending for Action', is a firm position on what 'closed litigation' means; it is undocumented, unconfirmed, and any change ...	Add a `\$Status` vocabularies table to docs/NOTICES_LITIGATION_FLOW.md listing every observed `gst_refund_applications.status` / `gst_drc03_filings.status` value with its Open/Closed mapping (pull the distinct values from the live tables with ...	S
G59	Medium	data-model	No staff workflow fields on refund or DRC-03 rows (owner, status, remarks, reply date) Nobody can be assigned a refund, mark it followed-up, or record that the RFD-09 reply went; the panel cannot show 'Replied' even though the reply exists.	Add the same workflow columns (staff_status, priority, assign_to, remarks, reply_date, next_action_date) to both tables, and derive replied from REPLY folder items when reply_date is null.	M
G60	Medium	data-model	Refund claim period, application status, bank account and RFD-04 flag are captured but have no columns Without the claim period the s.54(1) two-year limitation for re-filing after a deficiency memo cannot be computed, and provisional (RFD-04 90%) vs final refund cannot be distinguished.	Add period_from, period_to, application_status, rfd04_provisional boolean, bank_account_masked to `gst_refund_applications` and populate from the APLCN item in handleRefundDocs or from case/search itemJson.	S

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G61	Medium	data-model	Three different definitions of 'closed' for the same refund case The same case reads Closed in its folder, Open in the summary and unknown in the company profile; staff cannot trust any one screen.	Add a derived `case_state` (open / replied / sanctioned / disbursed / rejected / withdrawn / deficiency) computed once (Postgres view or shared util from the latest folder event) and have every page read it, with staff_status as an ...	M
G62	Medium	email-alerts	No communications log tying emails/calls/WhatsApp to a notice Litigation follow-up is communication-heavy; without a log the dashboard cannot show 'client not yet informed' or 'awaiting client docs for 12 days', and there is no defensible record that the client was told of a ...	CREATE TABLE gst_matter_communications (id uuid PK DEFAULT gen_random_uuid(), matter_id uuid NOT NULL REFERENCES gst_matters ON DELETE CASCADE, notice_id uuid REFERENCES gst_notices ON DELETE SET NULL, channel text NOT NULL CHECK (channel ...	S
G63	Medium	email-alerts	Failed portal logins are recorded but nobody is told — no sync-failure or stale-sync alert A client whose portal password changed silently drops out of monitoring; the team assumes 'no new notices' when in fact the pull never ran. The extension is manual and CAPTCHA-gated (extension/README.md:40-41 'It's not ...	DB trigger `trg_client_sync_log_failed` AFTER INSERT ON client_sync_log FOR EACH ROW WHEN (NEW.status='failed' AND NEW.action='notices') → insert gst_notice_events kind 'sync.failed' (dedupe_key `sync.failed:{client_id}:{date}`), picked up ...	S
G64	Medium	email-alerts	Sender cannot attach the notice PDF that the extension already captures A 'new notice' email that makes the reader log into the app to open the PDF halves its value; partners read on phones and need the notice inline.	Add `attachments jsonb` on email_outbox (`[{name, storage_path   url, content_type}]`). In send-gst-email, for each attachment download from the Supabase Storage bucket with the service key (or fetch the pdf_url), base64 it and append `{ ...	M
G65	Medium	email-alerts	Single recipient only — no cc/bcc, no team distribution list 'Email everything to the GST team' means a distribution list plus the assignee plus, on escalation, a partner — none of which can be expressed today, and the team list should be editable by the principal without a code ...	Add `cc_emails text[]`, `bcc_emails text[]` to email_outbox; in sendMail map them to `ccRecipients` / `bccRecipients`. New single-row table `firm_alert_settings` (id int primary key default 1 check (id=1), team_emails text[] not null ...	S

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G66	Medium	litigation-gap	Priority tiers exist but the Priority tile was removed in favour of Failed Logins Staff can tag High priority in the list view, but the dashboard never surfaces those tags, so the tag has no payoff and nobody maintains it.	Six tiles: Open, Overdue (destructive), Due in 7 days (warning), Unassigned (info), High priority, Demand ₹; each with a 7-day delta caption and 30-day sparkline. Total notices and GSTIN count move to the header subtitle; Failed Logins + ...	S
G67	Medium	litigation-gap	No counsel/advocate, authority or forum details Appeals and tribunal matters are argued by external counsel; the firm needs to know who holds the brief, their fee, and which authority/forum the matter is before to prepare the right form (APL-01 vs APL-05).	CREATE TABLE gst_counsels (id uuid PK DEFAULT gen_random_uuid(), name text NOT NULL, firm text, email text, phone text, bar_enrolment_no text, default_fee numeric, is_active boolean NOT NULL DEFAULT true); ALTER TABLE gst_matters ADD ...	S
G68	Medium	litigation-gap	No fees/billing hook per matter Litigation is separately billable from monthly compliance; without a hook the firm cannot see unbilled matters, counsel disbursements to recover, or fees quoted vs collected per dispute.	CREATE TABLE gst_matter_fees (id uuid PK DEFAULT gen_random_uuid(), matter_id uuid NOT NULL REFERENCES gst_matters ON DELETE CASCADE, fee_type text NOT NULL CHECK (fee_type IN ...	S
G69	Medium	litigation-gap	No reply-drafting workspace, templates or precedent library Replies to ASMT-10, DRC-01A, DRC-01, RFD-08 and appeal grounds are the firm's actual product; without drafts, versioning, standard paragraphs (16(4), 17(5), RCM, e-invoice) and past winning replies, quality depends on ...	Table matter_documents (matter_id, kind [notice, intimation, order, reply_draft, reply_filed, annexure, working, precedent, correspondence], version, file_url, uploaded_by, uploaded_at, filed_on_portal boolean, portal_ref) and ...	L
G70	Medium	litigation-gap	Counsel / external advocate management does not exist GSTAT/HC matters and many first appeals are argued by outside counsel; the firm needs who holds the brief, when the brief was sent, fee status and contact details per matter.	Master counsels (name, firm, forum expertise, phone, email, fee_basis) and matter_counsels (matter_id, counsel_id, role, brief_sent_on, fee_agreed, fee_paid, notes); show on the matter header and in the hearing record.	S
G71	Medium	litigation-gap	No litigation fee / cost tracking or time log per matter Litigation is billed separately from compliance retainers; without quote/billed/received per matter the firm under-bills reply drafting and hearings and cannot compute matter profitability.	Table matter_fees (matter_id, stage, quoted, billed, invoice_no, received, received_on) and optional matter_time_entries (matter_id, user_id, minutes, work_date, note); Fees tab on the matter and a 'Unbilled litigation work' report.	M

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G72	Medium	litigation-gap	Recovery and attachment proceedings (DRC-13/16/22, DRC-07A) have no structured tracking or stay status Bank-account garnishee (DRC-13) and provisional attachment (DRC-22) are emergencies with statutory objection windows (7 days for DRC-22A); the firm also needs to record that filing APL-01 with pre-deposit gives a deemed ...	Table <code>matter_recovery_actions</code> (<code>matter_id</code> , form [DRC-07A, DRC-13, DRC-16, DRC-22, DRC-22A, DRC-23], <code>issued_on</code> , <code>addressed_to</code> (bank/debtor), <code>amount</code> , <code>objection_due</code> , <code>stay_status</code> [none, <code>deemed_s107_7</code> , <code>court_stay</code>], <code>stay_reference</code> , <code>released_on</code>); ...	M
G73	Medium	litigation-gap	Git history was squashed on 2026-08-27, so the 2026-08-24/25/26 decisions cited in comments are unrecoverable from commit messages The comments are now the sole primary source; the moment an audit dimension rewrites those files, the dates, the Notice Alert comparisons and the scoping rationale vanish with no way to reconstruct them.	Before any refactor lands, transcribe every dated 'confirmed live' / 'decision' comment (list in the positions-register finding) into <code>docs/NOTICES_LITIGATION_FLOW.md</code> with its original date and file, so the doc becomes the primary record ...	S
G74	Medium	litigation-gap	Known data limitations that staff must interpret are recorded only in scattered comments (DRC-03 cash/credit split, unconfirmed CLOSURE code, blank Reply Type, always-empty 'Additional Notices' report, no reply automation) A GST-team member reading the DRC-03 Credit Ledger report or a case folder's empty Closure/Reply Type column has no way to know the blank is a documented limitation rather than a missing filing — and may act on an ...	Add a '\$Known limitations' table to <code>docs/NOTICES_LITIGATION_FLOW.md</code> (limitation → where it shows → what staff should do), and surface the two that affect figures (DRC-03 split payments, always-empty Additional Notices report) as an on-page ...	S
G75	Medium	portal-sync	No sync-run summary or completion signal; the dashboard never refreshes after Sync All Staff must switch to the portal tab to know where the run is, and must manually reload the dashboard to see new counts. A run that dies leaves no record that it ever existed.	Add <code>sync_runs(id, started_by, started_at, finished_at, mode, clients_total, clients_done, clients_failed, new_notices, aborted_reason)</code> written by <code>background.js</code> at start and by <code>content.js</code> on each advance/finish; add <code>run_id</code> to ...	M
G76	Medium	portal-sync	No retry/backoff for Supabase writes, PDF uploads or portal API errors, and no 'retry failed clients' path A transient 502 from Supabase during the delete+insert of a client's notices leaves that client with zero rows (delete succeeded, insert failed) until someone notices and re-runs a full firm-wide sync with 92 CAPTCHAs.	Wrap <code>sel/post/patch/del</code> in a retry helper (3 attempts, 500 ms/2 s/5 s backoff, retry on network error and 5xx/429); make the write transactional through the upsert RPC so a failed insert never follows a successful delete; add a 'Retry ...	S

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G77	Medium	portal-sync	Extension version is never checked by the app The extension is side-loaded ('Load unpacked', extension/README.md) on each staff PC and does not auto-update. Older builds still run older scrape logic (e.g. before the GSTR-3A Referer fix), so different staff PCs can ...	Include version in the announce message ({_gstkExtensionReady:true, version}) and in every client_sync_log row (new column extension_version); keep MIN_EXTENSION_VERSION in the app and show an 'Update the GST Keeper extension (you have ...	S
G78	Medium	portal-sync	Portal outstanding-demand / liability ledger is not captured alongside notices A DRC-07 demand that has crystallised in the liability register is a recovery risk (s.79) that the Recovery row of the summary table (currently 1) does not reflect.	Add an extension capture for the liability register Part-II (other than return) into `gst_demand_ledger (client_id, demand_id, order_no, period, tax, interest, penalty, status, pulled_at)` and surface an 'Outstanding demand ₹' tile plus a ...	L
G79	Medium	reporting	Notice Summary table: 14 placeholder dash rows push real data below the fold, Replied column is structurally always empty The most-used table on the page is 40% dashes and has a whole column that can never be non-zero for two of its biggest rows, which trains users to distrust it.	Sort real rows by Open desc, collapse placeholders behind a 'Show uncaptured categories (14)' toggle with a 'not captured' Badge, replace Replied with 'Awaiting reply' (open and no reply_date), draw an inline open/total bar per row, and ...	M
G80	Medium	reporting	Open (1060/1060) and Over Due (473) are inflated: null status counts as open, and portal closure signals (orders, closure items, LUT/ack rows) are ignored on the dashboard When every notice is 'Open' and half are 'Over Due' the tiles stop meaning anything and the team stops looking at them; genuine overdue replies are hidden in noise.	Add a computed system_status on gst_notices/matters set at sync from case-folder items (ORDRS / CLOSURE / DRC7A present → stage advanced), treat informational types (LUT, acknowledgements, RFD-11A) as non-actionable, and make Open = ...	M
G81	Medium	reporting	MIS reports limited to two: Notice Summary and GSTIN-wise count Partners need a monthly MIS (backlog, ageing, staff workload, demand exposure, outcomes) and clients need a per-GSTIN litigation status statement for their auditors; neither exists.	Add to the Report dropdown: Ageing (buckets x category), Staff Workload, Demand Exposure by Client (contingent liability), Stage/Outcome report, Per-client Litigation Statement (PDF via noticeFormatPdfExport.ts pattern). Add a monthly cron ...	M

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G82	Medium	reporting	No per-client 'Notice Status Report' PDF for partner/client; the only notice PDF exporter is a GSTR-9 reconciliation format Partners review litigation on paper/PDF in client meetings and clients ask 'where do my matters stand'; today someone reformats an Excel dump by hand.	Add src/utls/noticeStatusPdfExport.ts using jsPDF + autoTable + drawFirmLogo: per-client register (ref, type, section, issue, due, status, assignee, reply/order, demand, next action) with an open/closed summary header; buttons on ...	M
G83	Medium	reporting	Refund/DRC-03 closed heuristics miss live status strings; 'Replied' can never be non-zero for them although the portal already records 121 replies The Replied column shows dashes for every row even though the firm has filed 121 replies the portal can see; the closed/open split for refunds treats a sanctioned refund as pending.	Add 'sanction' and 'concluded' to the closed regexes (document the elected position), and derive replied = case has a REPLY folder item (join gst_case_folder_items on case_id) with reply_date auto-filled from reply.decdtls.dt.	S
G84	Medium	security	Only one coarse permission key governs all notice editing; no separate keys for assign, close, delete, or bulk An employee allowed to log their own replies is also allowed to bulk-close 1000 notices or reassign work; a manager cannot restrict closing to seniors. Viewing is open to every staff role (isStaffRole) which is fine, ...	Add keys manage_notice_assignments, close_notices, delete_notices, bulk_edit_notices in UserControlPage PERMISSIONS and AuthContext can*() helpers; keep edit_notice_status for logging replies/orders on notices assigned to the user.	S
G85	Medium	security	portal_jobs is writable by the anon key, so anyone can enqueue jobs that the service-role agent executes with client credentials; portal_sessions cookies never expire or rotate Live count shows 123 portal_jobs rows, so the queue is in use. Anyone with the public anon key can insert `{client_id, job_type:'PUSH_GSTR3B_SAVE', mode:'live', payload:{...}}` and, if the agent is running, it will log ...	Restrict portal_jobs INSERT/UPDATE to the same signed app session mechanism used for credentials (or an RPC `enqueue_portal_job` that checks a staff session), add `expires_at` to portal_sessions with a nightly purge and a `DELETE` after a ...	M
G86	Medium	security	audit_log is the only append-only table and is not used for any notice or client event The firm already has the right primitive for an immutable deletion log and is not using it where it matters most; an auditor asking 'who closed this SCN and when' or 'who deleted this client' gets no answer.	Write to audit_log from: client delete (module 'clients', action 'delete_client'), bulk status/priority changes in NoticeWorkflowListView (:435, :455), per-row status change (:404), AddNoticeDialog insert, and from the sync RPC (module ...	S

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G87	Medium	security	Sync All, Company List Sync/Fetch and the per-row Portal icon have no dedicated permission and gate inconsistently A firm-wide Sync All logs the extension into every credentialed client one after another and writes client_sync_log/gst_notices for all of them; a stray click by an article can start 90+ CAPTCHA-gated portal logins. ...	Add `sync_notices` (gates NoticesDashboardPage Sync All, CompanyListPage Sync + Fetch Company, ReportsPage per-report Pull) and put the portal-open icons (NoticeWorkflowListView.tsx:917-928, ClientsPage.tsx:411) behind ...	S
G88	Medium	security	export_data permission exists but is never enforced — every notices export and bulk PDF download is open to all staff The Notices-All export carries GSTIN, demand amounts, remarks and assign-to for every client (AllClientsNoticesPage.tsx:190-204) — a complete litigation book in one .xlsx. The principal's User Control screen advertises ...	Either call `canExportData()` around the four Export buttons and the two 'Download all visible PDFs' buttons (and pass it into NoticeWorkflowListView/EvidenceEventListView as a prop), or introduce `export_notices` as the ...	S
G89	Medium	security	Extension app bridge accepts commands from any *.vercel.app page and never checks who is asking — it logs in as any client using credentials fetched with the anon key The extension is the firm's automated hand on the GST portal. Because it trusts any page on the vercel.app wildcard and never learns which app user (or role) clicked, the permission model built in the app cannot reach ...	Narrow `matches` to the firm's exact domains (gst.vjdesai.com and the one vercel preview host); in appbridge.js reject messages where `e.source !== window`; have the app include `{ userId, sessionToken }` from vjdesai_user in every __gstk* ...	S
G90	Medium	security	Manual-notice PDFs are uploaded to a public bucket and exposed via getPublicUrl — readable by anyone with the link, no role required SCNs and orders contain demand figures, PAN/GSTIN and sometimes bank details. A public URL in an exported Excel (the PDF column is exported at AllClientsNoticesPage.tsx:203) or a forwarded email leaks the document to ...	Make return-pdfs private (or a new notice-pdfs bucket), store the object path instead of a public URL, and mint short-lived signed URLs on click via a small helper (createSignedUrl) that runs only for staff/client-owner; the exporter ...	M

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G91	Medium	security	Two documented CAPTCHA positions conflict: agent README forbids auto-solving on GSTN-terms grounds while the extension manifest reserves fastcaptcha.org for a future auto-solve path The firm's exposure position on automated portal access (and any third-party CAPTCHA service seeing client portal sessions) is a principal-level decision; today one folder says 'never' and the other has quietly ...	Record the firm's decision in docs/NOTICES_LITIGATION_FLOW.md \$Legal/risk (human CAPTCHA only vs. permitted solver, written client authorisation on file, credentials at rest) and either remove the fastcaptcha.org host permission or note ...	S
G92	Medium	ux	Company onboarding card does not belong on a notices dashboard and lists the 5 most recently created clients, unrelated to notices Half of the left column is spent on client-master admin that CompanyListPage already provides, while the clients with the worst litigation exposure are invisible. The critical Sync All precondition (extension missing) ...	Replace with 'Clients needing attention' (top 8 by overdue then open, columns GSTIN, Trade name, Open, Overdue, Last sync, owner); move Add/Import/Template to /notices-company-list; put Sync All in PageHeader actions and render the ...	M
G93	Medium	ux	Layout wastes width at 1920px and breaks rhythm: fixed 380x620 right card, dead space under Company/Calendar, 8px rhythm ignored At 1920px the four tiles stretch to ~330px each with a 24px number, the right card stays 380px, and the left column ends ~300px before the right one; at 1279px and below (most staff laptops with the sidebar open) the ...	12-column CSS grid with gap-6: tiles row (6 x col-span-2), queue col-span-8 + summary col-span-4 (auto height, min-h-0), charts 3 x col-span-4, attention-clients col-span-8 + agenda col-span-4; breakpoints lg → 2 columns, md → 1; adopt ...	M
G94	Medium	ux	No in-app notification bell — the Sidebar Bell is just the nav icon Email is the primary channel but staff who are already in the app should see 'New: 3 notices, 2 due tomorrow' without opening their inbox; it also gives a read receipt the partner can rely on.	Table `staff_notifications` (id uuid pk, user_id uuid null /* null = broadcast to all staff */, kind text, title text, body text, link text, notice_id uuid, created_at timestamptz default now(), read_by uuid[] not null default '{}') with ...	M
G95	Medium	ux	Sync progress and failures are only visible as a banner inside the portal tab; the dashboard offers no live status strip The person who pressed Sync All has to keep the portal tab in the foreground to type CAPTCHAs anyway, but everyone else on the team (and the principal looking at the dashboard) sees no indication that a sync is running, ...	Replace the 11px hint with a sync status card: extension state (with version), agent heartbeat state, current run progress from sync_runs (realtime), last completed run summary (ok/failed/new notices), and buttons Sync All / Retry failed / ...	M

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G96	Medium	ux	No column sorting — list is hard-wired to newest issue date first Partners want 'largest demand first' or 'earliest due first'; staff want 'by client' to batch replies. None of these are possible without exporting to Excel.	Add sortKey/sortDir state, clickable TableHead with ArrowUpDown icon, numeric sort for Amount of Demand and date sort for date columns via parseDateLoose; persist per-view sort in the URL (? sort=due&dir=asc).	S
G97	Medium	ux	Bulk toolbar covers only Status and Priority — no bulk assign, extended due date, export-selected, or email With 1060 rows, triage is done in batches (assign 20 ASMT-10s to one employee, push an extended due date after an adjournment, email a set to the team). Each is a 20-click job today.	Extend the selection toolbar with Assign, Set Extended Due Date, Add Remark, Email GST team (opens compose with selected refs), Export selected (Excel/PDF); keep the 'x selected' pill and add 'Select all N matching' across pages.	M
G98	Medium	ux	Logging a reply and closing the notice are two separate interactions; the Edit dialog cannot change status The most common flow ('reply filed today, mark it done') requires opening the dialog, saving, then finding the row again and changing the Select — the reason Replied stays empty is that the two-step flow is skipped.	Replace the single 12-field dialog with intent actions in the drawer: 'Log reply' (ref, date, attachment, auto-status reply_filed), 'Log order' (no, date, demand, attachment, auto-status order_received + prompt 'file appeal by <date+3 ...	M
G99	Medium	ux	'Submission' nav link works but only for manually keyed submission fields; case-folder REPLIES are never counted The portal already knows a reply was filed (RFD-09/DRC-06 in the REPLIES folder) yet 'Submission' and the Replied column stay empty unless staff retype the ARN — the 0 in Replied on the live dashboard is a symptom.	After replaceCaseFolderItems, have the extension patch the parent gst_notices row (reply_ref_number, reply_date, order_number, order_date from REPLIES/ORDRS items when null); alternatively compute 'replied' in computeNoticeSummary via a ...	M
G100	Medium	ux	Delete confirm dialogs give no counts, no impact preview and no non-destructive alternative Staff cannot tell the difference between deleting a duplicate GSTIN entered by mistake (0 notices) and deleting a 10-year litigation client (hundreds of orders); both show the same one-line prompt.	Before confirm, run a count query across the litigation tables and the storage prefixes and render 'This will remove 42 notices (17 open, 3 with orders), 6 DRC-03, 3 refunds and 51 PDFs'; require typing the GSTIN for any client with > 0 ...	S
G101	Medium	ux	Only the first captured document is reachable; 93 of 102 refund PDFs and all DRC-03 folder attachments are hidden Staff open the Company Profile or case folder to find an order that is already stored; from the firm-wide list the order is invisible.	Render a documents popover/count per row grouped by tab, link the ARN cell to /notices-case-folder/:clientId/:arn (EvidenceEventListView already supports this via clientIds), and fall back to the case-folder attachment when pdf_url is null.	S

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G10 2	Low	accessibility	Mobile: 23-column nowrap table and 12-field dialog are unusable at phone width Staff at a hearing or a client office check due dates on a phone; the current list needs sideways scrolling across ~2500px with 11px text.	Below md, render a card list (client, ref, type, due badge, status, assignee, one-tap Open drawer) instead of the table; make the drawer full-screen on mobile; collapse the filter row into a 'Filters' Sheet.	M
G10 3	Low	bug	Overdue/due-soon filters duplicated in four places with slightly different semantics A tile says 473, the drilled list's own filter says a different number after applying the dropdown; confidence erodes.	Centralise in src/utills/noticeDates.ts (effectiveDueDate, daysToDue, isOverdue, isDueSoon, isActionable) and have all four call sites import it; pass the pre-filtered record set to the view instead of re-deriving from formatted cells.	S
G10 4	Low	data-model	No tags or SLA fields on notices/matters The GST team cannot group matters by theme (e.g. 'ITC 16(4)', 'RCM on legal fees', 'Amnesty 128A') to reuse replies, and the dashboard has no basis for 'not looked at within 2 working days of receipt' escalation emails.	ALTER TABLE gst_matters ADD COLUMN tags text[] NOT NULL DEFAULT '{}', ADD COLUMN first_response_due timestamptz, ADD COLUMN first_responded_at timestamptz, ADD COLUMN escalation_level smallint NOT NULL DEFAULT 0, ADD COLUMN ...	S
G10 5	Low	data-model	claimed_amount is 0 for 10 of 37 rows and reason codes shown raw ('EXPWP'/'EXPWOP') Totals understate claims and the Refund Type column is unreadable for half the rows; 0 and 'unknown' are conflated.	Store claimed_amount NULL when ttlRfdAmt is absent (fall back to refClaim sum from the APLCN item) and map EXPWP/EXPWOP/ITC codes to display labels in one shared table.	S
G10 6	Low	email-alerts	No per-recipient preferences, quiet hours or opt-out; hourly send cron adds up to 59 min latency to urgent alerts A T-1 reminder that queues at 09:00 and sends at 09:59 is fine; a 'new SCN with 3-day window' that waits an hour is not, and partners will unsubscribe from a channel that pings them at 23:00 on a Sunday.	Add 'not_before timestamptz' to email_outbox and '.or('not_before.is.null,not_befor e.lte.now()')' to the sender query; table 'staff_notification_prefs (user_id uuid pk, email_new_notice bool default true, email_due bool default true, ...	M
G10 7	Low	litigation-gap	s.128A amnesty (SPL-01/SPL-02) matters lose their identity by being bucketed as 'Others' The waiver scheme is the single biggest settlement opportunity for s.73 demands of FY 2017-18 to 2019-20 (interest and penalty waived if tax paid by the notified date); the firm should see which open matters qualify, ...	Dedicated category 'Amnesty (s.128A)' with stages spl01_filed / spl02_filed / spl03_scn / spl04_reply / spl05_order / spl07_rejection; an 'eligible for 128A' computed flag on matters where section = 73 and FY in the scheme window and no ...	S

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G108	 Low	performance	Four ad-hoc effects fetch the whole gst_notices table client-side on every mount, with no caching or aggregate query Every tile click costs two full-table downloads (1000-row pages) and a re-render of 1060 rows; latency will grow linearly with notices and hurts the 'glance at the dashboard' use case.	One useQuery(['notices-dashboard', filters]) with staleTime 60s calling an RPC notice_dashboard_stats(fy, client, owner) that returns tile counts, category rows, ageing buckets and the top-N queue; keep fetchAllRows only for exports.	M
G109	 Low	performance	Missing indexes for the dashboard's actual access paths At 1,060 rows this is invisible; at 10 clients × 5 years of history it becomes sequential scans on every dashboard load, and case_id lookups from the folder page are already unindexed.	CREATE INDEX idx_gst_notices_source_status ON gst_notices (source, staff_status); CREATE INDEX idx_gst_notices_due ON gst_notices (due_date) WHERE staff_status IS DISTINCT FROM 'Closed'; CREATE INDEX idx_gst_notices_case ON gst_notices ...	S
G110	 Low	performance	Every DRC-03 sync re-downloads and re-uploads all PDFs and rewrites all rows Sync All over 92 GSTINs re-pushes hundreds of PDFs through the portal and Supabase storage each run, slowing the CAPTCHA-gated session and risking timeouts.	Read existing (arn → pdf_url) for the client first and skip download when present; upsert on (client_id, arn).	S
G111	 Low	portal-sync	client_sync_log has no run/job id and mixes diagnostic actions with sync outcomes A debug row can flip a client to 'Failed' on the Company List while the dashboard tile says 0, and there is no way to see 'this Sync All run covered 92 clients, 3 failed' as one unit or to count notices added/removed ...	ALTER TABLE client_sync_log ADD COLUMN run_id uuid, ADD COLUMN level text NOT NULL DEFAULT 'outcome' CHECK (level IN ('outcome','debug')), ADD COLUMN rows_seen int, ADD COLUMN rows_new int, ADD COLUMN rows_gone int, ADD COLUMN pdfs_ok int, ...	S
G112	 Low	portal-sync	Manual Add Notice requires hand-typing every field; no PDF/OCR extraction Offline notices (post, email from officer, IT/TDS) are the ones most likely to be mis-keyed and therefore mis-dated.	On file select in AddNoticeDialog, call an edge function `parse-notice-pdf` (pdf text extraction; regex for GSTIN, Ref No, DIN, dates, section, amount; fallback to Claude) and pre-fill the form for review. Low priority — portal-pulled rows ...	L

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G113	Low	portal-sync	No Income Tax / TDS notice coverage (all benchmark products span GST + IT + TDS) The firm's litigation exposure is not GST-only; a single dashboard is the pitch of every competitor. Out of scope for a GST product today but should be a known gap in the roadmap.	Keep the GST-only scope for now; design the new lifecycle/assignment/alert tables with a `tax_type` text discriminator ('GST' default) so IT/TDS notices (entered manually via AddNoticeDialog or a future agent job) reuse the same workflow, ...	XL
G114	Low	portal-sync	Operational constants of the sync job (20-minute expiry, 3x session-drop retry, 60-second CAPTCHA poll, per-client logout) are undocumented outside content.js With 92 credentialed GSTINs at 1–2 minutes each, a full Sync All cannot finish inside the 20-minute job window, so staff who start 'Sync All' from the dashboard will see the job silently stop mid-batch and have no text ...	State these constants and the batching consequence in the Sync All SOP (docs/NOTICES_LITIGATION_FLOW.md §Sync procedure) and in extension/README.md; consider raising startedAt-based expiry to a per-client inactivity timeout so a long ...	S
G115	Low	security	inject.js broadcasts every Blob created on any gst.gov.in page to all listeners with target '*' Every downloaded return/notice PDF is re-broadcast as a base64 data URL to any script in the portal page (other extensions, injected ads, a compromised portal asset). Low today because the portal is a trusted origin, ...	Post with `location.origin` as targetOrigin instead of `*`; include a per-job nonce that content.js sets in a DOM attribute before triggering the download, and have content.js verify `e.source === window`.	S
G116	Low	ux	Calendar widget conveys almost nothing: month tints without counts, no agenda, navigation on click The user wants 'what is due this week'; the widget makes them click a tinted day, leave the page, and read a list to learn it was a two-year-old closed notice.	Replace with a 'This week' agenda (7 rows: date, N due, N issued, first two client names, link) or keep the month grid with count badges and a Popover listing the day's open notices; compute markers only over open notices.	M
G117	Low	ux	Two category-row click behaviours look identical: some rows navigate away, others toggle a hidden local filter Users click DRC 03 expecting the tiles to filter (as Appeal did a second ago) and are thrown to another page; conversely they click Audit and do not notice the tiles changed.	Make the category label a filter chip action everywhere (adds a visible chip to the filter bar) and expose navigation only through the per-cell links; show the active category as a chip next to the KPI row with an X.	S

#	Priority	Area	Gap and why it matters	Proposal	Eff.
G118	Low	ux	Phone/tablet: 11px five-column tables and fixed-width GSTIN column overflow inside cards Partners check the dashboard from a phone between hearings; today the trade names truncate to a few characters and the summary needs horizontal scrolling inside a 620px-tall card.	Below md: render the summary as a stacked list (category, open/total bar, overdue chip), collapse the queue to card rows, drop the icon circle from tiles, and give any remaining table its own overflow-x-auto container.	S
G119	Low	ux	Outbox UI and email shell are hard-wired to return/period semantics and will mis-render notice kinds Once notice kinds exist, the only audit surface for what was emailed to whom shows blank return/period cells and the wrong letterhead accent, and the template editor's preview (EmailTemplatesEditor.tsx VARS/SAMPLE lines ...	RemindersPage: KIND_META map covering all kinds with icon/colour, replace 'Return · Period' with a 'Regarding' column (client name + notice reference or 'Firm digest'), add a Kind filter tab group and a Recipients column showing to/cc. ...	S
G120	Low	ux	No mobile push / app presence for alerts Partners check deadlines from the phone; a push on 'new DRC-01 for X, due in 30 days' is the cheapest alert channel after email.	Add a PWA manifest + service worker (vite-plugin-pwa) and Web Push (VAPID keys in Supabase secrets; `push_subscriptions` table); have queue-notice-alerts fan out to push in addition to email. Lower priority than email/WhatsApp.	L
G121	Low	ux	Company Profile KPI tiles are not clickable and lists are capped at 5 with no filters The page that should be the client's litigation home shows five rows and dead tiles; staff bounce to the firm-wide list for anything beyond that.	Wire each tile to /notices-all? client=:id&filter=...; replace the two 5-row panels with a full NoticeWorkflowListView scoped to the client (reuse buildViewNoticesAndOrdersReport) plus a client-level Activity feed and 'Email client status' ...	S
G122	Low	ux	User Control shows 22 flat checkboxes with the notices key buried last; no grouping by module and no explanation of role bypasses The principal will be adding six or seven litigation keys; without grouping (Clients & Credentials / Returns / Builder / Notices & Litigation / Administration) and a role banner, the screen becomes unreadable and ...	Add a `group` field to AVAILABLE_PERMISSIONS and render grouped Cards with a short group description; show an inline notice 'GST Manager already has every permission; these switches only apply to Employees' when selectedEmployeeData.role ...	S
G123	Low	ux	Stale error copy tells staff the DRC-03 pull does not exist Staff on the 84 clients without dedicated rows are told to enter records manually instead of running Sync All.	Change the message to 'Use Pull / Sync All to fetch DRC-03 filings from the portal' and update the migration comment.	S

B Appendix B · Competitive benchmark

Today the Notices Dashboard is a faithful clone of Notice Alert's *counting* layer (8 KPI tiles, a category table, a mini calendar, a company panel) but stops there: it has no alerting of any kind (the email pipeline in queue-gst-reminders only knows filing_status return reminders), no stage beyond Open/Closed, no owner, no hearing date, no demand-exposure figure, no charts, no staff-side document store, no activity log, and nothing a client can see. Every competitor verified (Notice Alert, ClearTax, Cygnet, GSTHero, Masters India, Turia, Jamku, GSTNoticeAI, My Notice Track) ships at least email alerts on new notice + due-date reminders and most ship owner/SLA workflow, hearing tracking and demand-at-risk views; several add WhatsApp, AI-assisted reply drafting and a client-facing status feed. The single biggest lever is wiring gst_notices into the existing email_outbox/pg_cron pipeline (new-notice digest to the GST team, 7/3/1-day due reminders, overdue escalation to the manager) — it reuses infrastructure that already exists and converts the dashboard from a scoreboard into an alerting system. Ranked features that would make this best-in-class for a mid-size CA firm: (1) email alerts + escalation to the GST team on new/due/overdue notices, with a daily digest; (2) a litigation lifecycle (Received → Reply Filed → PH → Order → Appeal → Tribunal/HC) with a case/matter grouping SCN, order and appeal; (3) named-staff assignment with a "My queue / Needs attention today" panel and staff-workload MIS; (4) demand-at-risk KPIs (tax/interest/penalty, pre-deposit, DRC-03 paid, PPR risk class) with a contingent-liability report; (5) hearing-date tracking on the calendar with hearing reminders and Google/Outlook .ics export; (6) per-notice activity log / audit trail and a staff document store (reply drafts, evidence, orders) on the case folder; (7) ageing/trend/category charts using the recharts kit already in the repo; (8) a client-portal "My Notices" view plus one-click "notify client" email with the PDF; (9) reply templates per notice form (ASMT-10, DRC-01A, REG-17, RFD-08) with an optional AI first-draft behind human review; (10) sync-freshness visibility and a scheduled agent pull so "Last 24 Hours" is not dependent on someone clicking Sync All and solving 92 CAPTCHAs.

Feature seen in the market	Evidence	How it fits this app
No email alert when a new notice is pulled or a due date approaches (every benchmark product has this)	queue-gst-reminders reads only <code>`from('filing_status').select('id, client_id, return_type, period_month, target_date, status')`</code> (line 224) and <code>`client_reminder_settings`</code> (line 214); <code>`email_outbox.kind`</code> is constrained to <code>`CHECK (kind IN ('reminder','confirmation'))`</code> (supabase/migrations/20260725120000_gst_reminders.sql line 47). A repo-wide grep for 'notice' across the reminders migrations, both edge functions, <code>gstReminders.ts</code>, <code>taskReminders.ts</code> and <code>RemindersPage.tsx</code> returns zero hits. Benchmarks: Notice Alert — "Upon successful downloading of a notice, the system can trigger notifications or alerts to the designated recipients" and "Due date reminder via email & Google / Outlook calendar" (https://www.microvistatech.com/gst-incometax-tds-notice-alert); GSTHero — "you will receive an email notification on new notice or due dates" (https://gsthero.com/gst-litigation-management-tool/); Masters India — "instant email notifications ... automatic alerts 7 and 14 days before deadlines" (https://www.mastersindia.co/notice-management-software/, via search summary); My Notice Track — "Detailed notifications sent instantly to your inbox the moment a notice is issued" (https://mynoticetrack.com/).	Extend the existing pipeline rather than build a new one: (a) widen <code>`email_outbox.kind`</code> CHECK to add <code>'notice_new','notice_due','notice_overdue','notice_digest'</code>; (b) add <code>`notice_id uuid`</code> to <code>email_outbox</code>; (c) new edge function <code>`queue-notice-alerts`</code> scheduled on the same <code>pg_cron</code> block (20260726130000_gst_email_cron.sql) that scans <code>gst_notices</code> where <code>pulled_at > last run</code> (new-notice alert to a firm-level <code>`notice_alert_recipients`</code> list stored in settings), <code>due_date/extended_due_date</code> in <code>{7,3,1}</code> days (reminder to assignee + GST team), and <code>due_date < today</code> with <code>staff_status != 'Closed'</code> (escalation to <code>gst_manager/superadmin</code>); (d) three editable templates in <code>`email_templates`</code> (merge fields: GSTIN, trade name, notice type, ref no, issue/due date, amount of demand, PDF link); (e) a 'Notice alerts' tab on <code>RemindersPage</code> to configure recipients and offsets. <code>send-gst-email</code> already drains the outbox hourly, so delivery needs no new code.

Feature seen in the market	Evidence	How it fits this app
No staff-side document store on a notice (reply drafts, evidence, working papers)	The only staff upload is the single attachment on manual add: <pre>`supabase.storage.from('return-pdfs').upload(path, file)` (line 64) stored as `pdf_url`; the edit dialog (NoticeWorkflowListView lines 1021-1099) has no file field;</pre> <pre>`gst_case_folder_items.attachments jsonb` (20260828160000 line 20) holds only portal-captured PDFs. Benchmarks: IRIS LMS — "store the complete trail of litigation from 1st Letter to Replies, Hearing Proceedings to Final Order" (https://irisgst.com/gst-litigation-management-system/, via search summary); Cygnet — "Documents and evidence are intelligently linked to each case" (https://www.cygnet.one/blog/key-features-of-gst-litigation-management-software/); GSTHero — "GSTHero archive all previous and current notices data" (https://gsthero.com/gst-litigation-management-tool/).</pre>	<pre>INSERT INTO storage.buckets (id,name,public) VALUES ('litigation-docs','litigation-docs',false); CREATE TABLE gst_documents (id uuid PK DEFAULT gen_random_uuid(), client_id uuid NOT NULL REFERENCES clients ON DELETE CASCADE, matter_id uuid REFERENCES gst_matters ON DELETE CASCADE, notice_id uuid REFERENCES gst_notices ON DELETE SET NULL, drc03_id uuid REFERENCES gst_drc03_filings ON DELETE SET NULL, refund_id uuid REFERENCES gst_refund_applications ON DELETE SET NULL, doc_type text NOT NULL CHECK (doc_type IN ('notice','intimation','reply','annexure','order','apl01','apl02','apl04','hearing_notes','drc03','refund_doc','client_communication','other')), title text NOT NULL, storage_bucket text NOT NULL DEFAULT 'litigation-docs', storage_path text NOT NULL, mime_type text, size_bytes bigint, sha256 text, portal_doc_id text, folder_section text, uploaded_by uuid, source text NOT NULL DEFAULT 'portal' CHECK (source IN ('portal','manual')), created_at timestamptz DEFAULT now(), UNIQUE (storage_bucket, storage_path)); CREATE INDEX ON gst_documents (matter_id); serve via createSignedUrl(path, 3600) in the app; migrate existing pdf_url/documents/attachments rows into it and move objects to the private bucket.</pre>

Feature seen in the market	Evidence	How it fits this app
Clients cannot see their notices and there is no one-click 'notify client'	<code>`if (!isStaffRole()) return <Navigate to="/dashboard" replace />`</code> (line 236; same guard in <code>CompanyProfilePage</code>, <code>AllClientsNoticesPage</code>, <code>AdditionalNoticeFolderPage</code>). <code>`grep -rni notice src/pages/DashboardPage.tsx`</code> returns nothing, so the client role has no notice surface at all, and no page has a 'send to client' action. Benchmarks: Turia — "Update clients over WhatsApp and email as responses progress" and "Every notice stays linked to its client" (https://turia.in/notice-management-software/); Suvit — "multi-client dashboards and automated communication tools, including WhatsApp reminders" (https://www.suvit.io/post/best-gst-software-2025, via search summary).	Seed template <code>`client_notice_intimation`</code> (kind <code>'client_notice'</code>, audience <code>'client'</code>): 'We have received <code>{{notice_type}}</code> <code>{{reference_number}}</code> dated <code>{{issue_date}}</code> for GSTIN <code>{{gstin}}</code>; reply due <code>{{due_date}}</code>; please share the following by <code>{{data_by_date}}</code>: <code>{{document_checklist}}</code>'. Add a 'Notify client' action on each row in <code>AllClientsNoticesPage</code> / <code>NoticeWorkflowListView</code> that opens a preview dialog (editable checklist per notice category from <code>noticeCategoryClassifier</code>), then inserts <code>email_outbox</code> {kind <code>'client_notice'</code>, <code>client_id</code>, to <code>clients.email</code>, cc team list, attachments <code>[pdf_url]</code>, <code>notice_id</code>, dedupe_key <code>'client_notice: {notice_id}:{n}'</code> and records <code>'client_notified_at'</code> on <code>gst_notices</code>. Never automatic — a CA must vet before the client is told.
Notice sync is manual, per-client CAPTCHA-gated and has no freshness indicator	<code>`handleSyncAll`</code> (lines 222-234) posts <code>`__gstkPullSectionAllClients`</code> to the extension and the success toast reads "Complete the CAPTCHA for each as it comes up." (line 208); agent/README.md lines 127-129: <code>"`login.ts` calls <code>`autoSolveCaptcha()</code> ... left unimplemented on purpose (returns null)". No 'last synced' timestamp is shown on the dashboard although <code>client_sync_log</code> exists (20260828100000). Benchmarks: Masters India — "auto-fetches notices and orders daily from GST, Income Tax, and Traces (TDS) portals" (https://www.mastersindia.co/notice-management-software/, via search summary); My Notice Track — "Automatic 24/7 tracking of GST and Income Tax portals" (https://mynoticetrack.com/); Cygnet — "pull notices as soon as they are issued, in real time" (https://www.cygnet.one/blog/what-is-litigation-management-system/).</code>	(a) Add a 'Sync freshness' strip on the dashboard: clients never synced / synced > 7 days ago / failed last attempt, computed from <code>`client_sync_log`</code> already fetched (lines 174-185), each linking to <code>/notices-company-list</code> pre-filtered; (b) show 'Last synced' per row on Company List and Company Profile; (c) a daily digest email listing stale/failed clients (part of <code>queue-notice-alerts</code>); (d) longer term, run the Windows agent's notices job on a schedule with a human CAPTCHA queue (agent/README already documents the scheduler pattern).

Feature seen in the market	Evidence	How it fits this app
No structured tagging by section, issue type, state or FY — dashboard filter is only Registration / Other	Type Of Notices select offers only `all`, `registration`, `other` (lines 340-342), and `financial_year` is a free-text Input (NoticeWorkflowListView line 1078); no `section`, `issue_tag` or `state` column exists in the migrations. Benchmark: Cygnet — "organizes them by GSTIN, state, financial year, and case status" and "Smart Issue Tagging ... classify disputes by department, statute, business unit, or issue type" (https://www.cygnet.one/blog/key-features-of-gst-litigation-management-software/).	Add `section_invoked` text`, `issue_tags` text[], `tax_period_from/to date` to `gst_notices`; derive `state` from GSTIN prefix (clients.gstin[0:2]) at query time. Replace the single select with a filter bar (FY, State, Section, Issue tag, Assignee, Priority) whose values propagate through `drillParams` so tiles, summary table and charts all honour it.
Portal outstanding-demand / liability ledger is not captured alongside notices	The firm-wide fetch reads `gst_notices`, `gst_refund_applications`, `gst_drc03_filings` only (lines 118-131); there is no table or extension handler for the Electronic Liability Register / outstanding demand (DRC-07) balance. Benchmarks: Jamku — "Manage and monitor outstanding demands effortlessly" (https://portal.jamku.app/); ClearTax — "Integrated access to returns, ITC ledgers, and invoices linked to notice periods" (https://cleartax.in/s/gst-notice-management-at-scale).	Add an extension capture for the liability register Part-II (other than return) into `gst_demand_ledger` (client_id, demand_id, order_no, period, tax, interest, penalty, status, pulled_at) and surface an 'Outstanding demand ₹' tile plus a Recovery-risk list on the dashboard.
MIS reports limited to two: Notice Summary and GSTIN-wise count	Report dropdown contains exactly two entries `/notices-report` and `/notices-gstin-wise-count` (lines 42-53). Benchmarks: Cygnet — "generate standard and custom reports by notice issue type, GSTIN-wise, region, or legal status" and "Audit-ready reports provide complete, end-to-end visibility into notices, responses, approvals, and outcomes" (https://www.cygnet.one/blog/what-is-litigation-management-system/ , https://www.cygnet.one/blog/key-features-of-gst-litigation-management-software/); GSTHero roadmap — "Reports and dashboards", "Litigation cost tracking" (https://gsthero.com/gst-litigation-management-tool/).	Add to the Report dropdown: Ageing (buckets x category), Staff Workload, Demand Exposure by Client (contingent liability), Stage/Outcome report, Per-client Litigation Statement (PDF via noticeFormatPdfExport.ts pattern). Add a monthly cron that emails the MIS PDF to superadmin/gst_manager via email_outbox.

Feature seen in the market	Evidence	How it fits this app
Manual Add Notice requires hand-typing every field; no PDF/OCR extraction	<code>`handleSubmit` validates `form.clientId    !form.refId.trim()    !form.type    !form.issuedDate`</code> (lines 46-49) — all keyed by hand; the attachment is stored but not parsed. Benchmarks: Cygnet — "Uses Intelligent Document Processing (IDP) to extract relevant fields—such as GSTIN, notice date, section, issue type, and due date—from uploaded PDFs" (https://www.cygnet.one/blog/what-is-litigation-management-system/); GST Notice AI — "OCR extracts all key fields: notice number, demand amount, GSTIN, tax period, due date, and officer details automatically" (https://gstnoticeai.com/).	On file select in AddNoticeDialog, call an edge function <code>`parse-notice-pdf`</code> (pdf text extraction; regex for GSTIN, Ref No, DIN, dates, section, amount; fallback to Claude) and pre-fill the form for review. Low priority — portal-pulled rows already carry fields.
No Income Tax / TDS notice coverage (all benchmark products span GST + IT + TDS)	All data comes from <code>`(q) => q.eq('source', 'notices')`</code> on <code>gst_notices</code> (line 149); the extension and agent target only the GST portal. Benchmarks: Notice Alert — "Auto Fetching Notice from GST, Income tax & TDS Portal" (https://www.microvistatech.com/gst-incometax-tds-notice-alert); Turia — "Tracks GST, Income Tax, and TDS notices" (https://turia.in/notice-management-software/); Jamku — "syncs data directly from Income Tax, GST, TDS portals" (https://portal.jamku.app/).	Keep the GST-only scope for now; design the new lifecycle/assignment/alert tables with a <code>`tax_type`</code> text discriminator ('GST' default) so IT/TDS notices (entered manually via AddNoticeDialog or a future agent job) reuse the same workflow, alerts and reports without a second module.
No mobile push / app presence for alerts	The app is a responsive web shell (docs/UI-UX-REDESIGN.md Phase 4: sidebar becomes a Sheet drawer below md); there is no PWA manifest, service worker or push subscription in the repo. Benchmarks: Notice Alert ships Android and iOS apps (https://play.google.com/store/apps/details?id=com.microvistatech.noticealert_final , https://apps.apple.com/us/app/-/id6702021191); My Notice Track — "Access your compliance dashboard and alerts on any device" (https://mynoticetrack.com/).	Add a PWA manifest + service worker (vite-plugin-pwa) and Web Push (VAPID keys in Supabase secrets; <code>`push_subscriptions`</code> table); have queue-notice-alerts fan out to push in addition to email. Lower priority than email/WhatsApp.

c Appendix C • GST forms referenced

Form(s)	Meaning
ASMT-10 / 11 / 12	Scrutiny notice, reply, acceptance
ADT-01 / 02	Audit notice, audit findings
DRC-01A / 01 / 02 / 06 / 07 / 08	Pre-SCN intimation, show cause notice, statement, reply, order summary, rectified order
DRC-01B / 01C / 01D	Liability mismatch, ITC mismatch intimations, recovery intimation
DRC-03 / 03A / 04	Voluntary payment, adjustment against demand, acknowledgement
DRC-09 / 13 / 16 / 20 / 22	Recovery: deduction, garnishee, attachment, instalments, provisional attachment
APL-01 / 02 / 04 / 05	First appeal, acknowledgement, appellate order summary, tribunal appeal
RFD-01 ... 09, PMT-03	Refund application through order, and re-credit
REG-17 ... 24	Registration cancellation SCN, reply, orders, revocation
GSTR-3A	Notice to return defaulter
MOV-01 ... 11	E-way bill inspection, detention, penalty, confiscation
SPL-01 / 02 / 05 / 07	Amnesty (s.128A) applications and orders
s.70 summons	Appearance before the officer

This document is a planning draft. Statutory references are to the CGST Act 2017 and CGST Rules 2017 as generally applied in 2026 and must be confirmed against the current text and notifications before being encoded as firm positions.

D Appendix D · Remaining medium and low defects

Verified defects of medium and low severity (20), continuing the numbering of §4.1, followed by the medium and low items from the follow-up reviews (9). All are scheduled with the phase that touches their area, most in Phase 0.6.

D.1 Verified, medium and low

#	Severity	Defect and impact	Fix	Eff.
B13	Medium	Partial failure inside the bundle is invisible: refunds/DRC-03 errors are not logged, and an unexpected exception stalls the job without a log row <code>extension/content.js:269</code> Company List shows 'success' for a client whose refunds and DRC-03 pulls failed, and the dashboard's Refund/DRC-03 rows in the Notice Summary silently go stale or get replaced by phantom rows. A thrown error in any handler (e.g. a Supabase 5xx) freezes the whole firm-wide run.	Give every step its own log row: action in ('login','notices','refunds','refund_docs','drc03'), plus run_id and duration_ms columns on client_sync_log; in the top-level catch write action=job.step status='failed' then advance(job). Company List status = worst status across the steps of the client's latest run; show a per-step ...	M
B14	Medium	NoticesTopNav shows 'Dashboard' as active on every page in the family <code>src/components/notices/NoticesTopNav.tsx:18</code> On Report > Notice Summary or Company List the pill group still highlights 'Dashboard', and the 'Notice' and 'Submission' pills are never shown active, so users lose their place inside the module.	Derive active from useLocation().pathname + search (matchPath for /notices-all with filter=submitted), or replace the pill group with the shadcn Tabs component in underline style and NavLink children.	S
B15	Medium	Inline status/priority edits revert on screen after any parent re-render <code>src/components/reports/views/NoticeWorkflowListView.tsx:210</code> Reproduction: on /notices-all mark a row Closed (toast says 'Status updated'), then change the Types Of Notices dropdown — the row shows Open again. The DB has 'Closed', so staff either re-do the edit or distrust the page; the KPI tiles on the dashboard also don't reflect the change until a full reload.	Wrap `table` in useMemo keyed on `filtered`, and pass an `onRowPatched(rowId, patch)` callback up so AllClientsNoticesPage updates `records` (or simply call fetchAll after a successful write). Same for the per-client report path that feeds this view.	S
B16	Medium	fetchAllRows pages without ORDER BY and swallows errors, returning partial data as if complete <code>src/lib/fetchAllRows.ts:22</code> Postgres offers no stable row order for LIMIT/OFFSET without ORDER BY; a Sync All running while the dashboard loads (delete + insert of a client's rows between page 1 and page 2, background.js:103-104) shifts the offsets so rows are duplicated or skipped, and the tiles show a wrong total with no error. A network/RLS error on ...	Append `.order('id', { ascending: true })` (or created_at,id) inside fetchAllRows before `.range()`, and either throw on error or return `{ rows, error }` so callers can toast and show a retry state instead of a silent partial count.	S

#	Severity	Defect and impact	Fix	Eff.
B17	Medium	Case folder page has no actions, no PDF viewer, and its Status disagrees with the dashboard's Open count <code>src/pages/AdditionalNoticeFolderPage.tsx:440</code> Two screens give two answers for the same case; staff reading the folder cannot act on it and must navigate back to the 23-column list to change anything; PDFs open in new tabs so cross-checking the notice against the reply is a tab-juggling exercise.	Derive portal_closed boolean at sync time (extension writes closure/order presence into <code>gst_notices.portal_status</code>) and let <code>isClosed</code> consider it; add the same action bar as the drawer on this page; embed an inline PDF viewer panel (iframe of the storage URL) beside the section tables; breadcrumb back to ...	M
B18	Medium	Company Dashboard lists and counts each refund/DRC-03 case twice <code>src/pages/CompanyProfilePage.tsx:193</code> Reproduction: open <code>/notices-company/<id></code> for a client whose refund ARN was captured by both the Refund tab sync and the case-folder sync — the ARN appears twice in 'Notices & Orders' and the 'Total Notices' tile is inflated by one per such case, while the firm-wide dashboard (which dedupes) shows a different number for the same ...	Build a Map keyed by case_id/ARN as <code>AllClientsRefundsPage</code> does (lines 79-92): let the dedicated-table row win, and only add a synthesized row when no <code>gst_notices</code> row with that case_id exists; apply the same to the Merged tab.	S
B19	Medium	Calendar 'Both' days render white digits on a pale amber tint - unreadable <code>src/pages/NoticesDashboardPage.tsx:498</code> Days that have both an issue and a due date are exactly the busiest days, and they are the ones whose number disappears. The day-2/day-7 tint on the screenshot is visible only because those days are single-marker.	Use text-foreground on all tinted days and mark categories with 4px dots below the number (info/destructive/warning) instead of background tints; or drop the month grid in favour of a 'This week' agenda list.	S
B20	Medium	No error state anywhere; Notice Summary empty state is unreachable dead code <code>src/pages/NoticesDashboardPage.tsx:545</code> If Supabase is down or the anon key is rejected, the page renders 0 notices / 0 overdue with full confidence. A partner cannot distinguish 'no work' from 'data failed to load'.	Return <code>{data,error}</code> from <code>fetchAllRows</code> , keep an error state per query, render an Alert variant=destructive with a Retry button at the top of the page, and use <code>TableEmptyState</code> (with icon) for genuinely empty queues.	S
B21	Medium	Icon-only buttons, calendar days and clickable table cells have no accessible names or keyboard access <code>src/pages/NoticesDashboardPage.tsx:484</code> Keyboard and screen-reader users cannot reach the summary drill-downs or understand the month buttons; it is also a direct breach of the firm's own enforced design rule.	<code>aria-label='Previous month' / 'Next month' / 'Edit \${c.name}';</code> day buttons <code>aria-label='13 September 2026, 2 notices due';</code> render cell links as <code><Link></code> elements (already used for KPI tiles) instead of <code>onClick</code> on <code><td></code> ; add <code>focus-visible:ring-2</code> to the Link-wrapped KPI Card.	S

#	Severity	Defect and impact	Fix	Eff.
B22	Medium	A notice is 'Over Due' from 05:30 IST on its own due date (UTC parse of date-only strings) <code>src/pages/NoticesDashboardPage.tsx:250</code> Reproduction: a notice due today (13/09/2026) shows in 'Over Due' for the whole working day while the reply can still be filed until midnight; it never appears in '7 Days Due' on its final day. Staff see a matter flagged late when it is still within time.	Compare date-only strings: const todayIso = local YYYY-MM-DD; overdue = due < todayIso; dueSoon = due >= todayIso && due <= todayIso+7d (compute the +7 by constructing new Date(y, m-1, d+7) locally). Put it in the shared noticeDueState helper.	S
B23	Medium	KPI tiles and Notice Summary use different definitions of a notice, so Total/Open never reconcile <code>src/pages/NoticesDashboardPage.tsx:252</code> Reproduction: add up the Notice Summary 'Total' column and compare with the 'Total Notices' tile on the same screen — they differ; likewise 'Open Notices' vs the sum of the Open column. A principal cannot answer 'how many open matters do we have' with one number from this page.	Create one canonical case set (a SQL view v_notice_cases or a shared TS builder) that: excludes PULL FAILED marker rows, collapses gst_notices rows by (client_id, coalesce(case_id, reference_number)), and unions dedicated refund/DRC-03 rows by ARN; drive tiles, summary, calendar and every drill-down from it. Also dedupe task ...	M
B24	Medium	gst_notices.updated_at is never maintained — no trigger, no updated_by <code>supabase/migrations/20260824100700_gst_notices_workflow_tracking.sql:8</code> There is no 'last touched' date to drive ageing, stale-item alerts or 'what changed since yesterday' emails to the GST team, and no record of who closed a notice or who assigned it.	CREATE TRIGGER trg_gst_notices_updated_at BEFORE UPDATE ON gst_notices FOR EACH ROW EXECUTE FUNCTION public.update_updated_at_column(); ALTER TABLE gst_notices ADD COLUMN created_by uuid, ADD COLUMN updated_by uuid, ADD COLUMN status_changed_at timestamptz, ADD COLUMN status_changed_by uuid; have the app pass user.id (from the ...	S
B25	Low	Sync All ignores the Company List's Active/Inactive flag <code>extension/background.js:396</code> Reproduction: mark a de-registered client Inactive via Bulk Update, then click Sync All — the extension still opens its login, demands a CAPTCHA and re-pulls its notices, and its rows keep feeding the tiles. The Inactive flag has no effect on anything but a display filter.	Add inactive_at_hand to API.getClients' select and skip 'c.inactive_at_hand' clients in startAllClientsSectionPull unless explicitly selected via info.clientIds; optionally exclude inactive clients' rows from the dashboard tiles by joining clients.inactive_at_hand.	S
B26	Low	Sidebar loses its active item on all Notices sub-routes <code>src/components/layout/Sidebar.tsx:168</code> Seven of the eight pages in the module show no highlighted sidebar entry; combined with the top-nav bug the user has no orientation cue at all.	Add an optional activePrefixes: ['/notices-', '/refunds-all', '/drc03-all'] to NavItem and compute isActive with location.pathname.startsWith in navLinkClasses.	S

#	Severity	Defect and impact	Fix	Eff.
B27	Low	Hooks are called after a conditional early return in three Notices pages <code>src/pages/CompanyListPage.tsx:142</code> If <code>isStaffRole()</code> ever changes while the page is mounted (permission snapshot refresh via <code>get_user_snapshot</code> changing the role, or <code>AuthContext setUser(null)</code> at <code>AuthContext.tsx:462</code> rendering before <code>MainLayout</code> swaps to <code><Navigate></code> at <code>MainLayout.tsx:30</code>), React throws 'Rendered fewer hooks than expected' and the page goes blank. It ...	Move the <code>if (isStaffRole())</code> return <code><Navigate .../></code> below all hook calls (or wrap the page body in a child component rendered only for staff).	S
B28	Low	Company card flashes 'No companies yet.' during load and hard-reloads the page after an import <code>src/pages/NoticesDashboardPage.tsx:447</code> Every visit shows a false empty state for a moment (on a firm with 92 GSTINs), and a successful bulk import throws away all page state and refetches the entire dashboard.	Add a loading skeleton (Skeleton rows) for the mini list; on import success refetch the list and show a sonner toast instead of reloading.	S
B29	Low	All dashboard queries ignore Supabase errors; a failed query renders as an empty firm <code>src/pages/NoticesDashboardPage.tsx:163</code> Reproduction: block the network or break an RLS policy — the tiles show 0 and the summary says 'No notices on record yet.' with no toast, indistinguishable from a clean database. Staff may conclude a client has no notices.	Capture <code>`error`</code> from each call, <code>toast.error</code> with the message, keep a <code>`loadError`</code> state that renders an inline 'Could not load notices — Retry' panel instead of the empty-state copy.	S
B30	Low	Calendar tints days from the unfiltered set but the day click carries the type/category filters <code>src/pages/NoticesDashboardPage.tsx:296</code> Reproduction: pick Type Of Notices = Registration, click a tinted day whose only notice is a DRC-01 — the drill-down opens an empty list, contradicting the tint.	Either derive <code>dayMarkers</code> from <code>displayRows</code> (so tint and drill agree with the on-screen filters) or drop type/category from the date drill; also add the 'closed' state to the markers once the due-state helper exists.	S
B31	Low	Import Company forces a full page reload and never refreshes after a partial import <code>src/pages/NoticesDashboardPage.tsx:411</code> Reproduction: import 10 companies where 1 row is invalid — 9 are inserted but the Company panel's 5-row list and 'Total GSTIN' stay stale until a manual refresh; when all succeed the whole app reloads, losing the selected Type/category filter and calendar month and re-running all five queries and the extension handshake.	Replace the reload with a <code>`loadMiniClients()`</code> function that re-runs the clients query (and, if desired, the notices query), and have <code>BulkAddClientsDialog</code> call <code>onSuccess</code> whenever <code>successCount > 0</code> .	S

#	Severity	Defect and impact	Fix	Eff.
B32	 Low	'Additional Notices and Orders' per-client report is a dead end: nothing ever writes source='additional_notices' <pre>src/utils/noticeRefundDrc03Reports.ts:33</pre> Reproduction: open Reports > Additional Notices and Orders for any client and click Pull — the report can never populate, yet its error text tells staff to Pull. The case-folder data that IS captured (gst_case_folder_items) is not surfaced by this report.	ALTER TABLE gst_notices ADD CONSTRAINT gst_notices_source_check CHECK (source IN ('notices','additional_notices','manual')); UPDATE gst_notices SET source='additional_notices' WHERE case_id IS NOT NULL AND notice_type IN ('Determination Of Tax','Letter Of Undertaking','Scrutiny Of Returns','Enforcement Case','Rectification Of ...	S

D.2 Follow-up reviews, medium and low

#	Severity	Defect and impact	Fix	Eff.
B55	 Medium	Partial case-folder or refund-document capture overwrites a previously complete one <pre>extension/background.js:143</pre> A single 20-second upload timeout (the code comments at background.js:229-235 describe a real 20-minute hang) silently shrinks a case folder that was complete last week — the ORDRS attachment link disappears from the Additional Notice Folder page even though the PDF is still in the bucket.	Merge instead of replace: key gst_case_folder_items on (client_id, case_id, folder_section, reference_number) with a unique index and upsert; when merging attachments, union the new list with the existing one by url. For refunds, carry the existing `documents` jsonb across the replace (or, after the RPC merge from the first ...	S
B56	 Medium	'Refund Claimed From Cash Ledger' report can never return rows; EXPWOP refunds misclassified <pre>extension/content.js:2910</pre> Two of the three refund reports the firm was promised are wrong or empty by construction.	Map refund reason codes explicitly (EXPWOP/ITC accumulated/inverted → ITC; EXPWP/excess cash/deemed export → Cash) in a lookup shared by the extension and the report, and store the raw code alongside the label.	S
B57	 Medium	Case-stub merge mislabels columns and picks an arbitrary event for multi-event refund cases <pre>src/pages/AllClientsRefundsPage.tsx:84</pre> The list reads as if applications were 'filed' on the day a rejection notice was issued, and which of three events is shown changes between page loads.	Model refund/DRC-03 as one row per case with an events array (from gst_case_folder_items) and display latest-event + latest-date columns; never coerce description into refund_type.	M

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B58	Medium	manage_employees lets an employee create GST Manager accounts, which have every permission implicitly <code>src/pages/ManageEmployeesPage.tsx:294</code> In the target litigation role model the manager tier must be a partner decision. Today an employee granted manage_employees can create a gst_manager login (default password 2026, ManageEmployeesPage.tsx:174) and thereby obtain every notices/clients/credentials right in the system, defeating the whole matrix.	Only superadmin may create/promote gst_manager (filter the role Select by user.role === 'superadmin'); manage_employees holders may create 'employee' only. Consider a distinct 'partner' bypass vs. gst_manager needing explicit grants for delete_clients, view_portal_credentials and delete_notice.	S
B59	Medium	CONFIRMED: 'Total GSTIN 92' means 'clients with at least one notice row', not GSTINs onboarded or synced <code>src/pages/NoticesDashboardPage.tsx:255</code> '92 of 121' hides the 13 credentialed clients that have never been synced at all — the exact clients whose notices the firm does not know about.	Show 'GSTINs 92 with notices / 121 onboarded / 13 never synced' and link the tile to /notices-company-list? status=never (the filter already exists at CompanyListPage.tsx L77-78).	S
B60	Medium	Stale, self-contradicting comments record the 'uncaptured categories' position that no longer holds <code>src/utils/noticeCategoryClassifier.ts:57</code> The comments are the only record of the 2026-08-24/25 scoping decisions, and they now say the opposite of what the code does; a reader cannot tell which categories are genuinely never pulled (and so need a portal reverse-engineering pass) from those that merely arrive via the case-type map.	In docs/NOTICES_LITIGATION_FLOW.md §Category taxonomy keep a two-column table: category → source (get/notices description regex case/task/get caseTypeName via CASE_NOTICE_TYPE_MAP dedicated refund/DRC-03 table not captured). Then trim the UNCAPTURED_NOTICE_CATEGORIES comment and the dashboard header comment to point at ...	S
B61	Medium	CONFIRMED: three different 'totals' on one screen — tile 1060, Notice Summary Total row 1106, GSTIN-wise 1090 <code>src/utils/noticeSummaryReport.ts:85</code> When the KPI tile and the table beneath it disagree by 46, every number on the page loses credibility with the person signing off.	Define one canonical unit ('matter' = distinct case_id for case rows, reference_number otherwise, ARN for refund/DRC-03) in a shared pure function and derive tile, table and GSTIN-wise counts from it; show the reconciliation ('1060 rows = 863 notices + 197 case tasks in 180 cases') as a tooltip.	M
B62	Medium	Closure regexes miss live portal statuses ('Proceedings Concluded', 'Drop proceedings') <code>src/utils/noticeSummaryReport.ts:52</code> Wording-based closure silently drifts every time the portal adds a status; each miss is a matter shown as pending that is not.	Replace the regexes with an explicit status → state lookup table (open / sanctioned / closed / withdrawn / deficiency) stored in code with a unit-like assertion over the live distinct values, and log unknown statuses to client_sync_log.	S

#	Severity	Defect and impact	Fix	Eff.
B63	 Low	CompanyListPage calls hooks after a conditional early return (Rules of Hooks violation) src/pages/CompanyListPage.tsx:142 If the same mounted component ever re-renders with isStaffRole() flipping (user object refreshed to null/client mid-session before MainLayout unmounts it), React throws 'Rendered fewer hooks than expected' and the Notices Dashboard's Company List goes blank. It also trips eslint react-hooks/rules-of-hooks, which hides real ...	Move the `if (isStaffRole()) return <Navigate .../>` line to just before the `return (` JSX at :248, after the last hook (:246).	S